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Review

The evolution of developmental biology through conceptual and technological revolutions

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SUMMARY

Developmental biology—the study of the processes by which cells, tissues, and organisms develop and change over time—has entered a new golden age. After the molecular genetics revolution in the 80s and 90s and the diversification of the field in the early 21st century, we have entered a phase when powerful technologies provide new approaches and open unexplored avenues. Progress in the field has been accelerated by advances in genomics, imaging, engineering, and computational biology and by emerging model systems ranging from tardigrades to organoids. We summarize how revolutionary technologies have led to remarkable progress in understanding animal development. We describe how classic questions in gene regulation, pattern formation, morphogenesis, organogenesis, and stem cell biology are being revisited. We discuss the connections of development with evolution, self-organization, metabolism, time, and ecology. We speculate how developmental biology might evolve in an era of synthetic biology, artificial intelligence, and human engineering.

INTRODUCTION

The field of developmental biology has made remarkable progress since the first issue of *Cell* was published in 1974 (see Table 1). While the 1970s were characterized by the description of intriguing phenomena and the identification of developmental mutants,[1,2] the molecular genetics revolution in the 1980s and 1990s led to the isolation of genes and pathways that regulate development,[3–28] revealed the remarkable evolutionary conservation of developmental mechanisms,[6–9,29] discovered the roles of developmental control genes in human disease,[12,30] and laid the foundation for elucidating fundamental mechanisms in developmental gene regulation, pattern formation, and organogenesis. The past 25 years have been characterized by an expansion of the field into areas ranging from morphogenesis, stem cell biology, and adult homeostasis to evolution and organ engineering. Recent breakthroughs in genomics, imaging, and genome editing technologies have led to global views of development and renewed interest in metabolism, developmental timing, ecology, and human development. Artificial intelligence (AI) and synthetic systems are likely to herald a new revolution in the field.

Here, we revisit some of the historical breakthroughs in the field, discuss the current status of developmental biology, and speculate as to what the future holds. We highlight how developmental biology is *in toto* biology that now covers the scale from molecule to ecosystem and has thus been at the leading edge of the life sciences.[137,138] We emphasize how developmental biologists have not only made important discoveries and conceptual advances but also developed revolutionary technologies that have been adopted in all of biomedicine (Table 1). We introduce powerful approaches that help elucidate the mechanisms of development, including model systems (Figure 1; Box 1), genetic manipulation and genomic recording (Figure 2), imaging (Figure 3; Box 2), synthetic engineering (Figure 4; Box 3), and theoretical (Box 4) and computational (Box 5) approaches. We begin with the classic topics of gene regulation, pattern formation, morphogenesis, organogenesis, and regeneration and then highlight recent progress that connects developmental biology to metabolism, timing, ecology, evolution, self-organization, and human development.

Attempting to cover 50+ years of developmental biology in one review is impossible. There are many alternative histories and views, but we hope that the presented examples help illustrate major concepts and technologies and highlight the current excitement and opportunities in the field. We attempt to be inclusive but focus on animal development and refer the reader to other sources that describe the fascinating development of microbes or plants, the remarkable specialization of the immune and nervous systems, and the intriguing developmental processes during reproduction. We apologize to our colleagues for omitting many important contributions and citations and encourage interested readers to turn to the cited textbooks, in-depth reviews, and the extended reference list (Document S1) for more information.

GENE REGULATION

Gene regulation during development is largely choreographed by transcription factors (TFs), *cis*-regulatory elements such as

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Table 1. Timeline of milestones since 1974

Advance	Concept vs. Technology	Reference
1970s		
C. elegans as a model system	Technology	31
Stem cell therapy	Technology	32,33
Regeneration by dedifferentiation and/or transdifferentiation	Concept	34–38
Transgenesis	Technology	39–41
Complete lineage of an animal	Concept	42,43
Genetics of pattern formation	Concept	1,2
Epithelial adult stem cells	Concept	44–46
Stem cell niche	Concept	47,48
In vitro fertilization	Technology	49
1980s		
Genetic saturation screens for developmental mutants	Technology	2
Zebrafish as a model system	Technology	50
Embryonic stem cells	Technology	51,52
Positional cloning	Technology	53
RNA *in situ* hybridization	Technology	54–56
HOX complexes, homeobox, and their conservation	Concept	3–9
Parental imprinting	Concept	57,58
Signaling pathways	Concept	10–14
Morphogens	Concept	59,60
Developmental genes are associated with human disorders	Concept	12,30
Master regulators	Concept	15–19
Enhancers	Concept	61–64
Extracellular matrix in development	Concept	65
High-resolution connectome	Technology	66
Cell sorting by cadherins and ephrins	Concept	20–22
Planar cell polarity and collective cell movements	Concept	23,67
Cell death pathway	Concept	24
Confocal microscopy in biology	Technology	68
Gene targeting by homologous recombination	Technology	69,70
Enhancer trapping	Technology	71
Cre/Lox, FLP/FRT, and GAL4/UAS systems for lineage, clonality, and gene function	Technology	72–74
Epithelial polarity proteins	Concept	25
Actomyosin in morphogenesis	Concept	75–77
Genetics of aging	Concept	78–80
Organoids	Technology	81–83
1990s		
Chromatin regulation of development	Concept	84
Clonal analysis and lineage tracing by stable and changing barcodes	Technology	85–87
MicroRNAs and long non-coding RNAs	Concept	27,28
Molecular nature of cytoplasmic determinant	Concept	26
Molecular nature of Spemann-Mangold organizer	Concept	88,89
GFP in animals	Technology	90
Mechanosensing in development	Concept	91–94
Segmentation clock	Concept	95,96
Cloning in mammals	Technology	97
Left-right patterning	Concept	98,99

(*Continued on next page*)

Table 1. *Continued*

Advance	Concept vs. Technology	Reference
First metazoan genome sequence	Technology	100
RNAi	Technology	101
Gene regulatory networks	Concept	102
Turing-like systems	Concept	103–106
Reprogramming and pluripotency in mammals	Concept	97,107
2000s to present		
Temporal patterning programs	Concept	108
Directed differentiation from embryonic stem cells	Technology	109,110
Molecular basis of guided cell migration	Concept	111,112
Light sheet microscopy	Technology	113
Morphological evolution through changes in regulatory elements	Concept	114
Large-scale regeneration by adult stem cells and lineage-restricted or multipotent progenitors	Concept	115–117
Rejuvenation factors	Concept	118
Induced pluripotent stem cells	Technology	107
In toto developmental live imaging	Technology	119
Self-organization from homogeneous populations or single cells	Concept	81–83,120
Whole transcriptome single-cell sequencing	Technology	121
Liquid-like condensates	Concept	122
Nanobodies in development	Technology	123
Organized embryo-like assemblies	Technology	124,125
Synthetic cell-cell communication	Technology	126,127
Embryonic gene expression trajectories	Technology	128–130
Genome therapy for hemoglobin and dystrophy disorders	Technology	131–135
Image-based trajectory reconstruction	Technology	136

Concepts and discoveries made by developmental biologists and technologies developed by developmental biologists (please note that many additional technologies have been used but were not developed by developmental biologists). These represent only a small selection of milestones in the field.

enhancers, and factors that modify and remodel chromatin.[178,179] These elements work in concert with cell-cell signaling to form complex gene regulatory networks (GRNs) that control the precise timing, location, and level of gene expression necessary for proper development.[102] The progress in the field initially relied on approaches such as genetic screens,[2] gene cloning,[53] RNA *in situ* hybridization,[54–56] and transgenic approaches[39–41,69,70,72–74] in genetically tractable model systems[1,2,31,50] (Figures 1 and 2; Box 1). These approaches revealed how cascades of TFs lead to fine-grained tissue patterning and cellular specialization. For example, the regulatory cascade from the Bicoid gradient to the segment-specific expression of Hox genes[3–9] revealed the logic of *Drosophila* early embryonic patterning and has become a paradigm for transcriptional regulation in development. Complementary cell culture approaches identified master regulators of cell type identity such as MyoD, which can drive muscle cell differentiation.[16] The development of reporter genes led to the identification of enhancers as *cis*-regulatory elements through which combinations of TFs generate precise spatial and temporal patterns of gene expression.[61–64,71] The discovery of different histone or DNA modifications led to the concept that the specificity of TFs is determined not only by their DNA-binding motifs but also by the chromatin context. These modifications, such as methylation and acetylation, can also act as a cellular memory, retaining a record of past transcriptional activities and influencing future gene expression patterns and responses to stimuli.[84] The integration of TFs, enhancers, and chromatin into GRNs has served as the blueprint for the unfolding of gene expression during development and has revealed network motifs such as feedforward and feedback loops that are repeatedly employed in developmental decision making.

These classic studies revealed fundamental concepts about developmental gene regulation but also raised new questions. What are the biophysical processes that influence developmental gene expression? What are the GRNs that confer cell-type- and tissue-specific expression? How does genome conformation affect developmental gene expression? And what are the signaling and mechanical cues that regulate the expression and activity of TFs?

On the smallest scale, some key areas of ongoing research are the biophysical mechanisms by which TFs interact with enhancers during development, how enhancers interact with promoters, and how mRNA production is regulated. Single-molecule RNA fluorescence *in situ* hybridization (FISH) and high-resolution live imaging of nascent RNAs revealed that

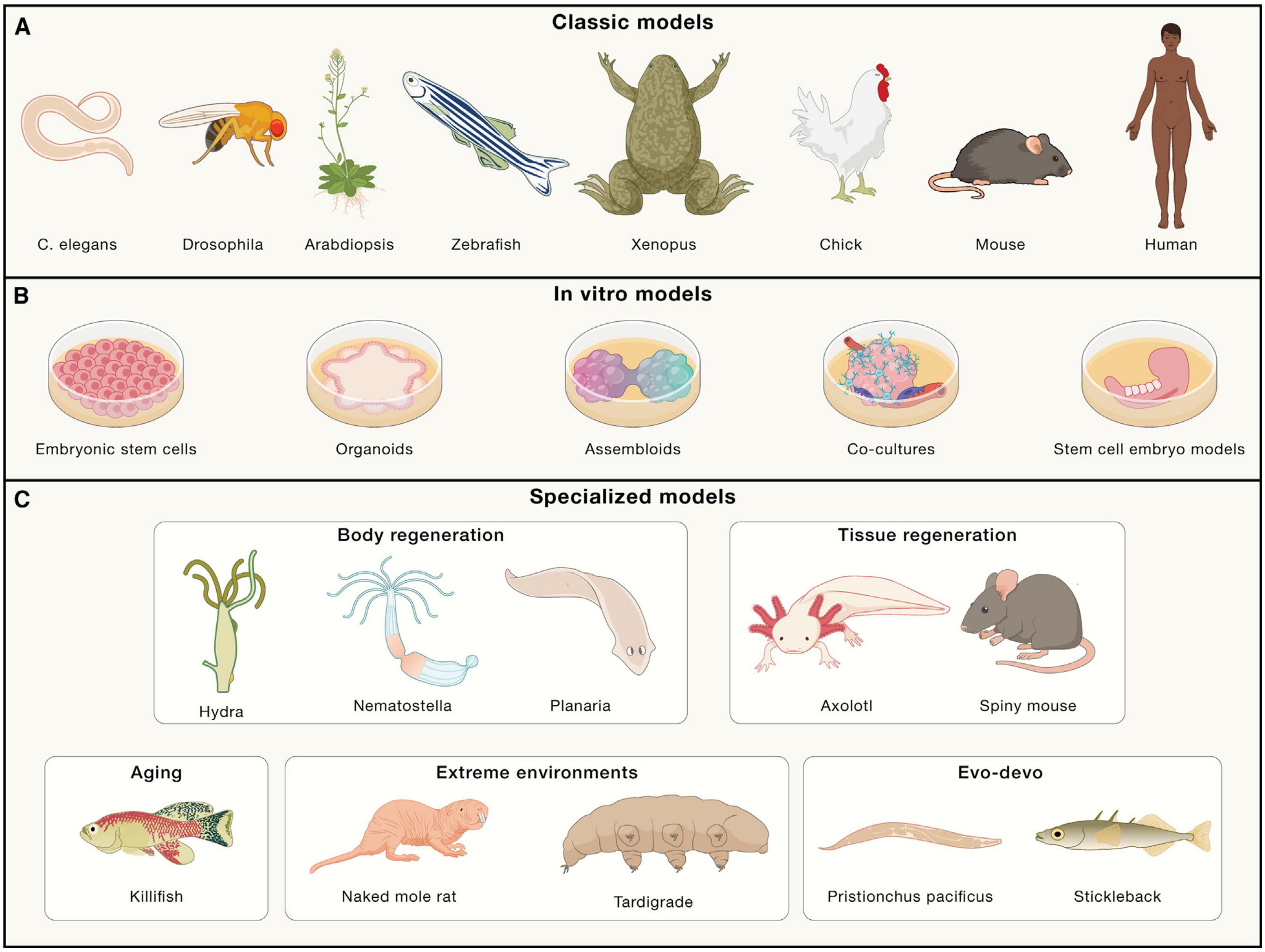


Figure 1. Models of developmental biology
(A) Classical models.
(B) *In vitro* models.
(C) Specialized models.

mRNAs are produced during transcriptional bursts[180] (Figure 3; Box 2). TF-enhancer interactions increase the frequency (and to some extent the duration and amplitude) of transcriptional bursts. Such bursts might lead to noisy gene expression, but temporal averaging through RNA and protein accumulation leads to more uniform expression. Tagging proteins, RNAs, and endogenous loci, as well as expansion microscopy and *in vivo* imaging (Figure 3; Box 2), support the concept that the formation of promoter-enhancer loops contributes to transcription.[178,179,181] However, such interactions can be short-lived and infrequent and might at times be indirectly mediated through "condensates" or "hubs" that promote transcription. Biophysical measurements have also shown that TF binding sites in developmental enhancers are often of low or medium binding affinity, making transcriptional activation dependent on combinatorial interactions and highly sensitive to TF concentrations, thus preventing ectopic gene activation.[182] Analogously, TFs can be suboptimal in their potency to activate gene expression, promoting the need for combinatorial control.[183] Improved *in vivo* imaging and structural biology methods promise to provide an integrated view of the steps from TF expression to enhancer binding to transcription in developmental systems.

At the genome-wide scale, the advent of new genomics technologies has revolutionized the field (Figure 2).[100] While initial studies of GRNs were hindered by the low throughput in mutating TFs, analyzing changes in target genes, or generating reporter genes, progress in high-throughput sequencing, genome editing, and single-cell analyses of RNA[121] and chromatin have provided global views of gene expression and chromatin during development.[102,128–130,184,185] We have entered an era in which cell types and cellular trajectories can be described at high molecular, temporal, and spatial resolution. Notably, even in the absence of perturbations, presumptive GRNs can be inferred by combining single-cell multiomics with binding site predictions.[186] These advances have resulted in the generation of detailed atlases that allow us to survey the complexity of developing tissues and the realization that hundreds of TFs regulate millions of enhancers, which regulate thousands of genes to

Box 1. Model organisms and systems in development

The study of development has relied upon model species. Sea urchin, salamander, frog, and chick laid the foundations to study morphogenesis. Molecular genetics in *Drosophila* and *C. elegans*[31] have enabled developmental gene discovery. Zebrafish became an important model due to its optical transparency and genetics.[50] Mice remain the paradigmatic mammal for studying development, including adult homeostasis and repair. *In vitro* studies were revolutionized by the derivation of mouse[51,52] and human[139] embryonic stem cells (ESCs). Building on insights gained *in vivo*, developmental factors are used to steer ESCs through stepwise specification toward specialized cell types. ESCs have also helped unveil the intrinsic self-organizing capacity and gene regulatory logic underlying embryonic patterning. Such knowledge is now being applied toward directing organoid self-assembly.

Stem-cell-derived organoids have launched an era of modeling mammalian development. Generally, organoids are classified into two categories: they are either derived from adult stem cells or from ESCs (or iPSCs). Following early studies of embryoid bodies,[120] early organoids recapitulated brain, pituitary, and optic cup development from ESCs.[83] Mammary-gland-like organoids were derived from adult stem cells[81] and revealed the important role of the ECM (e.g., matrigel). Single stem cells generated intestinal organoids.[82] While adult stem cells from diverse organs can self-organize into organotypic structures via regenerative processes, ESCs can spontaneously self-organize into many tissue organoids with strong similarities to *in vivo* development.[140] Organoids can be combined to generate assembloids, which contain multiple organoids or cultured cell types.[141] ESCs from mouse and human are also used to develop stem-cell-derived embryo models, such as 2D and 3D gastruloids,[124,125,142] ESC-derived embryos,[143,144] and blastoids.[145] These embryo models, following classic studies of frog animal caps, are currently experiencing a surge in technical advances and beginning to include non-canonical model species in stem cell zoos.

Specialized organisms provide distinct advantages for studying specific phenomena. For example, planaria, Hydra, and Nematostella possess exceptional regenerative capabilities. Axolotls regrow complete limbs. Spiny mice model scarless tissue regeneration. Killifish exhibit rapid aging, diapause, and self-organized axis formation. Cavefish adapt to extreme environments by alterations in morphology and metabolism. The naked mole rat is a model for aging and resistance to harsh environments. Tardigrades resist extreme environmental conditions. Several emerging model systems are used for evolutionary studies, including ascidians, stickleback, Parhyale, *Pristionchus pacificus*, and cichlids. These specialized models can yield mechanistic insights unavailable in traditional model organisms and build on technological advances such as high-throughput sequencing and genome editing. See extended reference list for citations (Document S1).

orchestrate development. Examples have shown how atlases can lead to fundamentally new insights: for example, single-cell analysis of the thymus has revealed that organ and cell type regulators drive the expression of liver, neuron, muscle, and other programs to help eliminate self-reactive T cells.[187] Analyses of bifurcation points in trajectory trees suggest that lineage-determining TFs are initially co-expressed to maintain progenitors but then become mutually repressed to determine cell fate.[185] In some cases, a determined cell fate requires the continuous repression of alternative programs to maintain its identity.[188] The convergence of genomics and deep learning has helped predict endogenous TF occupancy and enhancer activity and brought us to the cusp of defining and designing regulatory elements *in silico*.[189] For example, training neural networks with accessible chromatin sequences, TF expression, binding site predictions, and target gene expression has allowed the computational reconstruction of developmental GRNs and the design of tissue- or cell-type-specific enhancers during embryogenesis.[186,190,191] Although gene regulation is inherently context-dependent, it is conceivable that future approaches will be able to use only genome sequence to identify (and design) elements that drive expression in specific cell types, time windows, or patterns during development. To test these predictions, massively parallel reporter assays and perturbation studies will need to be transferred from cell culture to *in vivo* systems.

At the scale of whole-genome structure, techniques such as chromosome conformation capture and multiplex FISH have revealed that the genome is organized into domains and compartments that regulate gene expression. These structures range from topologically associating domains (TADs), which restrict enhancer-promoter interactions to specific genomic regions, to chromosome-scale domains, which repress or promote gene expression.[192] For example, in the Hox complex, DNA loops and gene expression emerge and spread from one part of the cluster, creating a developmental timer,[193] and non-coding RNAs can regulate chromatin over vast distances to inactivate one X chromosome in XX females.[27,194] Chromosomal *trans*-interactions such as transvection or transcription hubs suggest additional complexity in genome conformations.[195] Exciting progress in genomics and imaging approaches has the potential to reveal if, when, and how genome folding regulates gene expression and how chromosome conformation might contribute to epigenetic memory.[196]

As the field of developmental gene regulation continues to evolve, several key areas present open questions and exciting future directions. How is specificity in TF-DNA binding generated? Remarkably, the function of some TFs can be replaced by TFs from a different class. Might specificity rely on the combination of many weak and to some extent non-specific interactions?[197,198] Can structural predictions help decipher the binding preferences of TF complexes?[199] How is precision and robustness in gene expression achieved? The use of multiple enhancers has been shown to contribute to robust gene expression,[200,201] but what are the underlying biophysical mechanisms? More generally, how important are the exact levels of gene expression?[57,58] In traditional model systems, most genes are not haploinsufficient, at least for severe phenotypes, but in humans, phenotypes can be exquisitely sensitive to the levels of a gene product.[202] Sophisticated approaches that tune expression levels will be needed to address this question.

Another key question is whether there is an underlying logic to the regulatory interactions in GRNs. It is clear that some TFs can act as master regulators to generate specific organs, cell types, cellular structures, or functions,[15–19,203–205] and the cross-regulatory interactions between TFs to generate and maintain

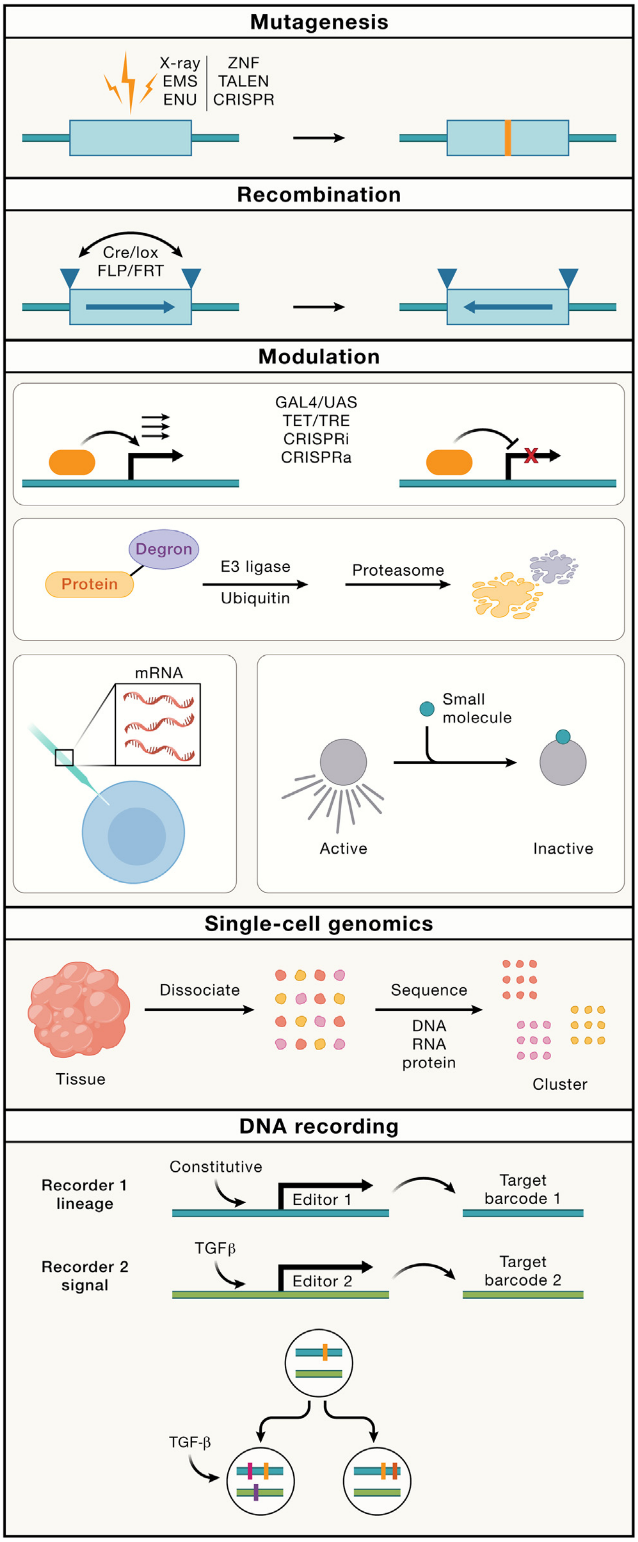


Figure 2. Molecular technologies of developmental biology

Elucidating gene function in developmental biology.

Mutagenesis: mutagens such as X-rays, ethyl methanesulfonate (EMS), ethylnitrosourea (ENU), zinc finger nucleases (ZNF), transcription activator-like effector nucleases (TALEN), or clustered regularly interspaced short palindromic repeats/Cas proteins (CRISPR) introduce changes in the genome, which might lead to phenotypic changes.

Recombination: the Cre/lox and FLP/FRT systems are used to conditionally activate or remove DNA elements. Applications include the generation of conditional mutants or the long-term labeling of groups of cells to determine what they give rise to (clonal analysis).

Modulation of gene activity: GAL4 and TET bind to upstream activating sequences (UAS) and TET-responsive element sequences (TRE), respectively, to activate (or repress) a target gene. Ectopic expression of GAL4 or TET in different tissues can activate the target gene ectopically. CRISPRi and CRISPRa program Cas9 and related proteins to repress or activate, respectively, genes targeted via guide RNAs. mRNAs can be synthesized *in vitro* and delivered into developing systems for mis- or overexpression. Proteins can be fused to degrons that are ubiquitinated by E3 ligases, followed by proteasome-mediated degradation. Small molecules can be used to inhibit (or activate) specific proteins.

Single-cell genomics: tissues are dissociated and the content (DNA, RNA, protein) of individual barcoded cells is analyzed by sequencing. Cells with similar features cluster together.

DNA recording: recorders consist of a DNA editor and a target locus containing a barcode that can be mutated by the editor. The activity of the editor can be constitutive (editor 1) or depend on intrinsic or extrinsic cues (editor 2). In the example, a cell contains recorders 1 and 2. Editor 1 is active and introduces a semi-random mutation (yellow) in target barcode 1. This mutation will be inherited by the descendants of this cell. Editor 1 is also active in both daughter cells but introduces semi-random mutations that are different (pink in one cell and orange in the other cell). One daughter cell receives a TGF-β signal, which activates editor 2, which induces a mutation (purple) in target barcode 2. Single-cell sequencing of target barcodes 1 and 2 reveals that the two cells are related (via the shared mutation [yellow]) and that one cell received a TGF-β signal (mutation [purple] in target barcode 2).

expression patterns and cell types have been studied extensively. But the targets of most TFs are not known—is there a logic to their regulatory roles, maybe akin to the regulation of bacterial operons, where a set of co-expressed genes is responsible for a cellular sub-function or sub-structure[206]? Similarly, what distinguishes seemingly different classes of TFs? Pioneer factors have the ability to remodel chromatin,[207] but what is the relationship and specificity of terminal differentiation TFs, organ-identity TFs, temporal TFs, signaling TFs, or the many remaining poorly studied TFs? The combination of global developmental atlases, large mutant collections, and structural studies might help reveal recurring patterns that help answer this question.

Finally, how do GRNs evolve, and what makes them adaptable to environmental changes? Many micro-evolutionary changes in development occur at the level of *cis*-regulatory elements that change the location or timing of gene expression (heterotopy and heterochrony), while changes in protein coding regions are less frequent. For example, the study of pigmentation patterns or skeletal morphologies has identified *cis*-regulatory changes that underlie evolutionary diversity.[114,208] However, we still have very few well understood examples for how genome evolution results in large-scale morphological changes (see "Evolutionary developmental biology"). The combination of genomics and AI might accelerate progress in the field. AI approaches have the potential to help decipher the grammar that links DNA sequence to TFs and sites of gene expression (Box 5) and might be able to predict from sequence alone how gene expression differs between two species. Understanding the evolution and evolvability of GRNs will provide insights into the diversity of life forms and their developmental strategies.

The development of synthetic biological systems offers another powerful tool for understanding gene regulation (Box 3).

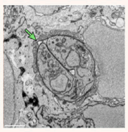
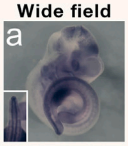
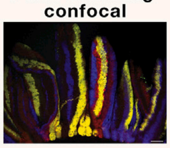
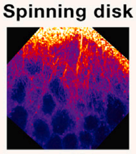
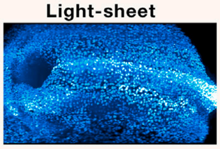
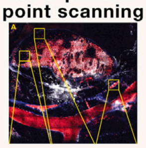
	Spatial scales	Current main application	Advantages	Main limitations
Electron microscopy 	Thin slices limit z for TEM For SEM up to mm sized volumes	Spatial organization of ultrastructures (extracellular matrix fibers, contact sites, organelles)	Molecular resolution of intracellular compartment and molecular complexes	No live imaging
Super resolution 	Can go down to < 5 nm in special cases	Structures below the diffraction limit of light (nuclear pores, centrioles, cytoskeleton, vesicles, cell protrusions, promoter-enhancer interactions)	Single molecule resolution	Very long acquisition times
Wide field 	From subcellular structures to large samples, <1μm - ~500μm	Imaging of 2D fixed sections or single molecule tracking in living cells	Sensitivity and speed	Out-of-focus light can dramatically reduce contrast
Point scanning confocal 	From subcellular structures to large samples, <1μm - ~500μm	Dynamic and subcellular to tissue-level information (gene expression patterns, protein localization, cellular arrangement, morphological features), protein life time, interactions and signaling dynamics (FLIM, FRET, FRAP, FLIP)	3D imaging of fixed large samples	Low imaging speed (single-beam scanning), lower sensitivity of detectors, bleaching due to higher laser power
Spinning disk 	From subcellular structures to tissue scales like organoids or embryos reaching several hundred micrometers	Fast imaging of subcellular to tissue-level information including cytoskeletal and junctional dynamics, cell division, migration, cell sorting and gene expression patterns, protein localization, cellular arrangement, morphological features, interactions and signaling dynamics (FRET, FRAP, FLIP)	Fast 3D imaging	Fixed pinhole size, i.e., fixed thickness for optical sectioning
Automated systems: wide-field/spinning disk confocal 	Reflects properties of wide-field / spinning disk confocal	High content and throughput phenotypic characterization (compound and genetic perturbation screens)	Sampling hundreds or thousands samples simultaneously	Large data storage required
Light-sheet 	From subcellular structures to whole organisms, <1 μm to ~1 mm sizes	Longterm imaging of large ex vivo/in vitro samples (organoids) and micron- to millimeter-scale model organisms (zebrafish, *Drosophila* and mouse embryos, *Hydra*)	Long-term live imaging of large samples	Lower resolution than confocal setups Sample mounting
Multi-photon point scanning 	Can resolve subcellular structures of entire tissues, from <1 μm to ~1 mm sizes	In vivo deep tissue imaging of embryogenesis, organogenesis and tumorogenesis (including microenvironment)	Deeper imaging in scattering tissues	Low imaging speed (single-beam scanning), cross excitation of fluorophores and bleed-through between channels

Figure 3. Imaging technologies of developmental biology

Properties of imagining approaches in spatial scales, current applications, and main limitations. Images from top to bottom: EM, gondii tachyzoite cells (green) inside the hindbrain of a zebrafish larva (TEM, transmission electron microscopy; SEM, scanning electron microscopy);[146] super resolution, SIM (structured illumination microscopy) image from time-lapse movies of actin dynamics in the growth cone;[147] wide field, whole-mount *in situ* hybridization of

(legend continued on next page)

Box 2. Probes and sample preparation for imaging

Recent years have witnessed dramatic progress in the development of microscopy approaches[113,119] and of probes that can be visualized, quantified, and perturbed at cellular and subcellular resolution. The advances have also included different types of bio-mechanical properties that can be measured and extracted (see text).[154,155] Since the cloning of GFP in 1992 and its transfer into animals,[90] many new fluorescent proteins have been developed through genetic mutation of photoproteins. Fluorescent tagging enables the *in vivo* imaging of expression and localization dynamics over time, also possible with self-labeling probes such as HaloTags. More recently, nanobodies[123] have emerged as *in vivo* detectors. These small antibody fragments are derived from camelid heavy chain antibodies and can be fluorescently labeled to serve as high-affinity tags targeting a protein of interest. Nanobodies can also be used in functional assays by trapping or degrading proteins. Biosensors to measure protein activity or mechanical[156] and metabolic states can follow the dynamics of cellular activities.[157] RNA visualization has undergone remarkable progress.[54–56] Single molecule RNA FISH (smFISH) offers major advantages in precisely localizing and quantifying numbers of many individual RNA molecules at once within single cells. This single molecule precision made smFISH an important tool for the more recent developments in spatial transcriptomics and led to the development of seqFISH and MERFISH. These spatial transcriptomics techniques map mRNA expression in tissues at single-molecule resolution and large scale (thousands of mRNAs).[158] The hybridization-based approaches are complemented by sequencing-based approaches that use array-based spatial capture methods or sequencing *in situ*. Continuing advances in labeling, capture, and computing are revealing a more detailed and comprehensive view of cellular tissue organization. For live imaging of RNA dynamics, the MS2 system,[159] which involves tagging RNA with MS2 loops and fluorescently labeled MS2 coat proteins, and the SunTag system,[160] which amplifies the fluorescent signal to a single RNA molecule, allow real-time observation of RNA transcription and localization. In summary, new genetically encodable fluorescent tags, RNA detection strategies, and minimally invasive live-cell probes have expanded the toolkit enabling visualization of molecules and cells.

Novel clearing techniques have been an important step to deal with large samples and the opaque nature of biological tissues. These limitations have been overcome by using chemical treatments to homogenize the refractive indices of tissues and cellular components. Methods like CLARITY, DISCO, PACT, CUBIC, and others employ detergent extraction of lipids combined with optical index matching solutions to render tissues transparent or tune their refractive index to adapt to various conditions.[161] This approach minimizes scattering while preserving fluorescent labels and fine architecture for whole intact organ imaging. Meanwhile, expansion microscopy (ExM)[162] physically expands tissues to increase the spacing between molecules before imaging. By swelling the polymer nearly 10-fold isotropically, structures previously below the optical diffraction limit can be resolved on a conventional microscope while maintaining molecular spatial context. See extended reference list for citations (Document S1).

By building synthetic GRNs and utilizing genome editing technologies, researchers can dissect the principles of gene regulation and create models to test hypotheses about developmental processes. For example, the synthetic MultiFate system can generate a scalable, combinatorial ensemble of stable gene expression states by engineering TFs that self-activate as homodimers and inhibit one another as heterodimers.[168] The *in vivo* relevance of such approaches remains to be fully established, but the artificial systems have the potential to reveal general principles of regulatory interactions.

Understanding developmental gene regulation has also been of medical relevance. For example, the study of erythropoiesis and hemoglobin gene regulation during development has led to the discovery of a transcriptional repressor (BLC11A) that blocks the expression of fetal hemoglobin in adults.[131,132] Strikingly, CRISPR-mediated mutation of a BCL11A enhancer reduces BCL11A expression, reactivates fetal hemoglobin, and can suppress symptoms of sickle cell anemia in humans. This approach has recently become the first approved CRISPR therapy and highlights the medical potential of understanding developmental gene regulation.

While this section is focused on gene regulation at the transcriptional level, post-transcriptional controls also play fundamental roles in many systems. This extends to all cells but is most apparent after fertilization, when the zygotic genome is not yet transcribed. Post-transcriptional mechanisms include RNA editing and localization, the production of multiple protein variants through alternative splicing or translational heterogeneity, and the regulation of RNA stability and translation through RNA-binding proteins and microRNAs.[28] It is unclear how all these elements work together to regulate the quantity and activity of gene products. For example, 5′ and 3′ UTR sequences are important determinants of mRNA stability and translation. While some sequences and RNA-binding proteins have been defined that mediate these regulatory functions, the interplay of these elements in gene expression is poorly understood. Studies that combine massively parallel reporter assays and AI will help decipher the regulatory code of UTRs and alternative splicing.

Translational control and post-translational modifications represent a critical and final level of gene regulation, profoundly impacting the expression and functional properties of proteins during development. Post-transcriptional regulation forms a complex web of controls that fine-tune gene expression during development, but our understanding of these interactions is rudimentary. For example, the correlation between RNA and protein levels is still poorly understood in most developmental systems. While changes in RNA levels correlate with changes in protein levels, the steady-state relationship between RNA and protein has been difficult to decipher. Moreover, modifications can alter protein activity, localization, stability, and interactions, providing a dynamic means to rapidly adapt the proteome to the changing needs of developing organisms. Large-scale and sensitive single-cell proteomics technologies are needed to help reveal the extent of translational regulation

wild-type embryos at E10.5;[148] point scanning confocal, R26R-Confetti mice small intestine (FLIM, fluorescence lifetime imaging microscopy; FRET, fluorescence resonance energy transfer; FRAP, fluorescence recovery after photobleaching; FLIP, fluorescence loss in photobleaching);[149] spinning disk;[150] automated systems, human colon intestinal organoid actin staining;[151] light-sheet, long-term live imaging of post-implantation mouse development;[152] multi-photon point scanning, distribution of vessels and cells within lymph nodes.[153]

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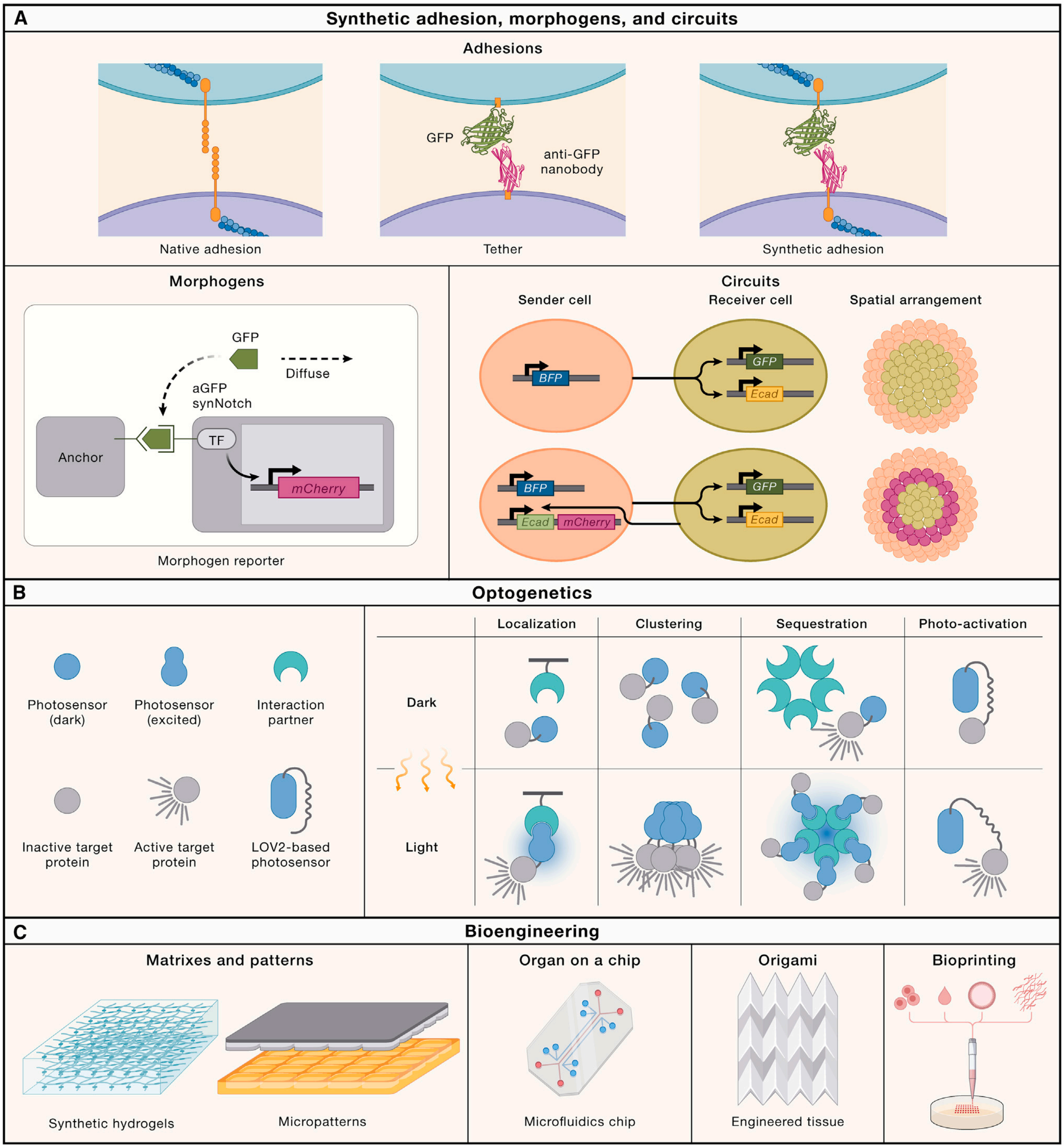

Figure 4. Bioengineering technologies of developmental biology

(A) Molecular components as modules to reprogram cell-cell communication with tethers and synthetic adhesion, morphogens with fluorescent reporters also in the presence of anchor cells, and decision making via genetic circuits changing the expression of E-cadherin that regulates patterning of cell populations.

(B) Optogenetic tools inducing cell patterning, lateral inhibition, sorting, migration, and tissue folding via intracellular protein changes in localization via binding to interaction partners in different localizations such as the plasma membrane, protein clustering to activate the protein, sequestration and inactivation of the protein, or uncaging small molecules or proteins to activate them.

(C) Controlling cellular behavior by changing matrix composition and generating engineered micropatterns in 2D and 3D; microfluidics connecting different organs on a chip; generating engineered patterned ECM origami that then shape epithelium in different shapes; or bioprinting cells, adhesion molecules, organoids, and ECM components.

Box 3. Synthetic developmental biology

Rational programming of spatial patterning has been a vision in developmental biology and bioengineering for decades.[163] Through the precise engineering of genetic circuits and the manipulation of biological systems and their environment, complex developmental processes are beginning to be controlled and modeled with high spatial and temporal precision.[164] These approaches can be grouped into three main areas: (1) building synthetic gene circuits to control cell-cell adhesion and cell fate; (2) using optogenetics to control cellular and tissue behavior; and (3) building engineered ECMs to control the environment.

In vivo studies have led to models for pattern formation (e.g., differential adhesion, morphogen shuttling, or lateral inhibition). Synthetic developmental biology has reconstructed, pared down, and tested these models. In these approaches, molecular components serve as modules to reprogram cell decision making and cell-cell communication to generate higher-order physiological functions and patterns.[165] Cell-centric engineering focuses on remodeling how cells sense and transmit signals. For example, several studies have developed modular toolkits to recreate cell-cell connectivity via orthogonal extracellular signaling systems[126,166,167] or synthetic receptors such as synNotch and synCAM.[127,165] These reagents enable tailored reprogramming of cells to user-defined inputs. Finally, engineered gene circuits act as programmable developmental regulators to control cell fate decisions in precise spatial patterns.[168]

Optogenetic tools induce developmental processes such as cell patterning, lateral inhibition, sorting, migration, and tissue folding. These minimal systems allow testing to discover which components are sufficient to drive a process outside of its native context. Optogenetics have also been applied in more complex systems to perturb and rescue developmental processes.[169]

A final important aspect of controlling cellular behavior is controlling the environment with synthetic ECMs.[91] These studies revealed that mechanical factors are important developmental cues. With new polymeric biomaterials and hydrogels, engineers have created designer synthetic ECMs and scaffolds that recreate properties of stem cell niches and generate boundaries to control cellular behaviors.[170,171] These approaches use customizable biochemical, mechanical, and topographical cues that provide microenvironmental signals to guide stem cell fate and morphogenesis. Overall engineering approaches have become central to the elucidation of the biophysical and spatial dynamics that orchestrate embryogenesis, organ formation, and regeneration, enabling complex organoid formation, bioprinting, and organ-on-a-chip devices. See extended reference list for citations (Document S1).

in development, and progress in *in vivo* imaging is required to clarify the dynamics of translation and its control.

Finally, additional complexity arises from the often exquisitely specific localization of RNAs and proteins in specific cellular locations and from the interaction of RNAs and proteins in non-membranous organelles ("condensates," "granules," "clusters," and "droplets").[26,122,209–211] It is still mysterious how these structures are assembled, display specificity, or function *in vivo*. A major challenge for the field is to develop technologies that specifically interrogate the role of condensates without disrupting other functions of the associated RNAs or proteins.

PATTERN FORMATION

Developmental gene regulation confers specific identities to individual cells, but these cellular programs need to be coordinated within a tissue and organism through the process of pattern formation. During pattern formation, groups of cells acquire distinct identities and arrange to form complex tissues and organs. These coordinated cell fate decisions are often mediated by intercellular signaling molecules.[212,213] Initial progress in the isolation of such signals and their antagonists and transduction machineries came from a convergence of approaches, including genetic and molecular screens (Figure 2) and biochemical purifications in model organisms and cultured cells (Figure 1).[10–14,88,89] These studies revealed both remarkable simplicity—a handful of signaling pathways are repeatedly used in most species and tissues (e.g., TGF-β, Notch, Hh, Wnt, FGF/RTK, NF-κB/dorsal)[214]—and extensive complexity—most signaling steps, from biogenesis to response, are highly regulated, modulated, and combinatorial. Some of these signals can act as morphogens—they function at a distance from their source of expression and can generate concentration-dependent effects, often in the form of gradients.[59,60,215–217] These initial studies raised fundamental questions: How are morphogens transported? How do cells interpret morphogens? Are there other means of patterning beyond morphogens?

Somewhat ironically, the first developmental morphogen to be identified was not a secreted intercellular signal but a TF, *Drosophila* Bicoid.[59] Conceptually similar to a signaling molecule, Bicoid is produced from localized mRNAs and then diffuses through a syncytium to form a concentration gradient. This mode of local production and long-range transport has also been shown for several secreted signals: a signal is produced locally, moves through effective diffusion that is hindered by binding proteins and trafficking, and is degraded. Imaging approaches have measured the transport modes of morphogens and have supported diffusive mechanisms[103,218–220] (Box 4). However, there are several additional mechanisms contributing to the distribution and activity of signals. The shape, morphogenesis, and growth of tissues can influence the geometry of signal distribution. Signals can be transported (and detected) by cellular extensions such as filopodia or cytonemes or trafficked intracellularly.[221] They can be transmitted through relay mechanisms, in which the signal acts at short range to activate its own expression or the expression of a secondary signal in neighboring cells. Inhibitors, receptors, extracellular matrix (ECM) proteins, and chaperones can shape the distribution of and response to signaling molecules. Regardless of the specific cellular mechanism that determines their distribution and transduction, all signals serve a common purpose: to provide positional information to cells within a tissue.[215,222,223] Such information can be in the form of morphogen gradients that define different positions, or it can be short-range and lead to lateral induction or inhibition. Some interactions can generate more complex patterns. For example, local autoregulation and long-range inhibition[103,104–106,224] can generate Turing patterns of stripes and dots,[225,226] positive feedback propagation can create

Box 4. Modeling approaches

Modeling approaches have become indispensable tools in developmental biology, enabling quantitative modeling and simulation of dynamic processes. Computational methods allow integration of large-scale datasets in genetics, genomics, cell behaviors, biophysics, and molecular interactions to generate predictive models of development. Key applications include mathematical models of enhancer-promoter interactions; data-driven approaches such as tracing cell lineage origins (see text), understanding the dynamics of signaling systems, or mapping gene regulatory networks (see "Gene regulation"); as well as predictive models of tissue patterning (see "Pattern formation") or tissue mechanics and collective behaviors that transform simple structures into complex organs (see "Morphogenesis" and "Tissue mechanics").[172] For example, by incorporating cell forces and tissue material properties, continuum models can capture aspects of limb bud shape changes during elongation, invertebrate whole-embryo patterning, and tissue-level transformations such as heart looping or brain folding. At another scale, agent-based models simulate individual cell behaviors to predict the emergence of cell movement coordination during processes such neural crest migration, and vertex models help simulate tissue structure, packing, and order.

Quantitative models evolved to better capture realistic anatomy, for example, predicting vertebrate limb bud shapes. Later, simulations combining cell shape changes, division, adhesion, and migration gave rise to predictions of gastrulation-like collective behaviors. Also in that period, the birth of computational systems biology started enabling genome-wide elucidation of developmental networks based on accumulating genetic and molecular data.[102] The development of novel methods to measure tissue mechanics as well as quantitative cellular tracking and multi-scale models have now also revealed how changes in material properties can organize morphogenesis.[173] Recent foci include organoid growth mechanics,[174] single-cell regulome networks (see "Gene regulation"), and machine-learning-enabled lineage tracing and the inference of differentiation trajectories (see "Gene regulation"). Currently, AI-guided simulations attempt to uncover emergent morphogenetic principles linking subcellular protein self-organization to cell cooperation at the multicellular scale, revealing how molecular-level variance canalizes into deterministic anatomical patterns seen across embryos of a species.[175] The combination of interpretability from biophysical models and the predictive power of machine intelligence opens new horizons in understanding development. See extended reference list for citations (Document S1).

traveling waves, and temporal feedback inhibition can generate oscillations.[95,96,223]

Just as morphogen transport is regulated at multiple steps, signal transduction is influenced by multiple factors. While some morphogen transduction is relatively linear, feedback mechanisms can amplify or dampen signaling, leading to non-linear outputs such as all-or-none responses or adaptation. In addition, the duration and location of signal reception influences signaling output. At the same concentration, longer ligand exposure can lead to higher pathway activity than short-term exposure, and the localization and compartmentalization of ligands and receptors can restrict and buffer signaling. These interactions can lead to complex signaling dynamics that do not lead to simple concentration-dependent linear effects but can have outputs that depend on signal concentration integrated over time or the interpretation of temporal fold changes rather than absolute levels.

The ultimate outcome of morphogen signaling is the differential regulation of target genes in space and time. In the simplest scenario, target gene expression is determined by the affinity of the morphogen-transducing TF for enhancer chromatin. High TF levels lead to the activation of enhancers with high- and low-affinity binding, whereas lower levels lead to activation of only high-affinity enhancers.[227] Importantly, affinity is not determined solely by DNA binding site affinity but by the combination of cofactors, binding sites, and chromatin accessibility. These interactions lead to the activation of short- and long-range target genes. However, following these regulatory steps, more complex patterns can be generated by the cross-regulatory interactions of target genes or pre-patterns in target tissue. For example, cross-repressive transcriptional networks can generate sharp striped expression domains. Thus, the "simple" graded morphogen input can generate "complex" patterning outputs.[227] Indeed, detailed measurements in the *Drosophila* anterior-posterior patterning system have revealed remarkable precision in the positional information generated by local concentrations of TFs.[228]

Despite the progress in the field, some fundamental assumptions have not been tested in depth. First, spatial gradients are a powerful way to generate patterns, but it has actually rarely been directly demonstrated that they are absolutely necessary. The Bicoid gradient can be greatly flattened without completely disrupting patterning, and some ligands (e.g., Wnt) can be tethered to the membrane to curtail their diffusion without major developmental defects.[229] In an extreme view, the spatially graded distribution of morphogens is simply the byproduct of local production of secreted signals but is not absolutely necessary for patterning per se. Technological advances in controlling the exact spatial and temporal expression of morphogens will be needed to address this question.[164] Reduced systems such as *in vitro* patterned tissues or organoids (Figure 1), as well as the development of optogenetic methods to rapidly localize protein activity to different regions (Figure 4; Box 3), will contribute to these efforts.

Second, signal diffusion or signaling relays are powerful ways to spread signaling, but effective diffusion is often limited to 1–20 cell diameters, and relays can take hours or days to spread. To rapidly pattern large structures, additional ways of signal spreading have been suggested. Cell bodies themselves have been shown to act as conduits ("superhighways") for trigger waves. For example, during wound healing in planaria, ERK waves can transverse long and parallelly aligned muscle fibers at 1 mm/h.[230] In contrast, Erk waves travel over days during the osteoblast regeneration in zebrafish scales.[231] These examples show how cell shape and arrangement influence signaling spread. Mechanical or electrical signals can also act quickly over large distances, but it remains to be determined how widespread such mechanisms are for pattern formation.[232] Improved imaging technologies that can follow the spread of signals and their transducers will be needed to distinguish between different models of signal transport.

Third, another fundamental problem in pattern formation is the phenomenon of scaling and how it is coupled to tissue growth,

Box 5. Computational and AI approaches

Computational methods have contributed extensively to the analysis of complex datasets in genomics (see main text) or imaging. The field has evolved from quantitative imaging in the 1970s to computational image enhancements that enabled measurements of geometric and intensity features from living cells and tissue sections. Algorithmic developments in image segmentation and edge detection built fundamental techniques for extracting structural information from biological forms. Conversion to digital data then increased acceleration, miniaturization, and throughput. At the turn of the millennium, high-content screening systems became capable of not just macro-imaging tissues or organisms but also imaging cells in complex tissues quantitatively. This high-resolution, high-frequency imaging of molecular dynamics and cell behaviors in model organisms along with massive genomic data laid the groundwork for modern AI in biology. Powerful deep convolutional and generative neural networks now exhibit human-like pattern recognition.[176] Trained on vast imaging datasets, deep learning can classify cells, track dividing cells in embryos, reconstruct lineages, and predict molecular interactions or anatomical transformations solely from images with remarkable accuracy. Future advances will involve real-time *in vivo* tracking of spatiotemporal developmental dynamics in 4D space using AI[177] and might help uncover principles underlying self-organization from collective cell behaviors. Moreover, representation learning has the potential to aggregate imaging and genomics information across scales into a common information space. See extended reference list for citations (Document S1).

wherein the same proportions of cell types are generated in tissues of different sizes. One prominent model for scale-invariant patterning postulates that the morphogen is coupled to accessory molecules that can inhibit (if tissue is smaller; contraction-induction) or enhance (if tissue is larger; expansion-repression) signaling.[233] However, alternative mechanisms might be at play in different tissues.

Fourth, how precise is patterning? Because of noise, a simple gradient can confer precision at high levels of morphogen but not at lower levels. In this case, counter-gradients or target gene interactions can create precise patterning.[228] For example, the opposing Hh and BMP gradients can set up a precise coordinate system during spinal cord development.[234] Additional mechanisms for precision include feedback regulation, temporal integration, and target gene interactions. Similar mechanisms might also confer robustness to genetic or environmental challenges, but the precise molecular mechanisms are still poorly understood. The combination of modeling approaches with the use of specific probes and tools to follow protein dynamics will accelerate progress in this field (Box 2).[220,223,228,235–237]

Synthetic approaches have been an additional strategy to extract the design features of signal-generated patterning[164] (Figure 4; Box 3). For example, tethering Wnt demonstrated the potential of short-range signaling in pattern formation,[229] using an inhibitory drug to replace the Nodal antagonist Lefty showed that feedback primarily serves to buffer signal fluctuations,[238] and using cultured cells to introduce the components of the Hh or BMP systems reproduced many of the observations seen *in vivo* such as receptor-mediated feedback inhibition or morphogen shuttling, respectively.[126,166] Finally, optogenetic approaches have been used to control the shape and timing of morphogen signaling.[164] For example, optogenetic activation of Nodal signaling helped reveal the roles of signaling duration and strength in patterning zebrafish embryos. In the future, the combination of such synthetic approaches with organoid approaches will help reduce complex *in vivo* networks to core interactions and create artificial patterns in the long run.

MORPHOGENESIS

The emergence of developmental patterns coincides with morphogenesis, the process by which an organism takes its shape. A key feature of this process is the dynamic interplay of decisions at the level of individual cells and their coordination with morphological changes at the large-scale tissue level.[239] Thus, the study of morphogenesis necessitates *in vivo* visualization at scales ranging from molecular components and single cells to whole tissues and even entire organisms (Figure 3; Box 2). It also requires sophisticated biomechanics measurements and perturbations,[154,155] such as micropipetting aspiration, laser ablation, hydrostatic pressure measurement, optical tweezers, and non-light-based imaging techniques such as Brouillon microscopy, (photo-) acoustics, and ultrasound imaging. In the following sections, we explore how these technologies helped reveal how the communication among individual cells regulates sorting and migration and examine the role of the ECM in coordinating cellular behaviors. We then discuss the roles of planar and apical-basal cell polarity. The roles of mechanics, forces, and osmotic pressures on tissue rearrangements will be addressed in the following section ("Tissue mechanics").

A major conceptual advance in our understanding of adhesion in morphogenesis was the differential adhesion hypothesis and the discovery of adhesion molecules. Classic studies revealed the spontaneous self-organization of amphibian embryonic cells *in vitro*: cells sort into different domains according to their adhesive properties. The differential adhesion hypothesis inspired the search for adhesion molecules, leading to the discovery of cadherins.[20,21] The use of atomic force microscopy and micropipetting aspiration to quantify the adhesive and mechanical properties of cells revealed an important role for cortical tension and actomyosin contractility in cell sorting.[240] Together with parallel investigations of repulsion mechanisms by non-adhesion molecules such as Eph/ephrin[22,241] and previous hypotheses on differential interfacial tension,[242] the current model postulates that differences in adhesion and cortical tension between cell populations can drive tissue organization. Synthetic approaches in cell culture have extended this model, identifying intracellular domains as the determinants of interface morphology and mechanics, while extracellular interactions determine the connectivity between cells.[165] Adhesion molecules are absolutely required, since they transmit contractile forces between cells, but there is still little evidence that adhesion molecules regulate adhesion strength per se in other ways than by signaling to the cytoskeleton. Taken together, these studies suggest that the crosstalk between extracellular-adhesion-mediated intracellular signaling and cortical tension regulates the sorting of cells.

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Beyond adhesion, improved imaging and biophysics approaches revealed the importance of cortical tension and actomyosin dynamics in cell rearrangements (Figure 3; Box 4).[75–77] For example, the implementation of faster spinning disks for live time-lapse microscopy revealed that actomyosin dynamics regulate junction remodeling during axial elongation and apical constriction during ventral furrow formation in *Drosophila*. But how are these mechanisms integrated with morphogen-mediated patterning? Progress in engineering of mosaic embryos, creation of synthetic tools, and improved microscopy techniques have begun to address this question. For example, in the zebrafish spinal cord, the initially noisy interpretation of a morphogen gradient leads to the scattered specification of different cell fates. Robust patterning and cellular arrangement are achieved through cell sorting because the differently fated cells express different cadherins.[243]

Cell migration is another key aspect of morphogenesis, driven by a combination of intrinsic cellular properties and extrinsic signals. Cells move alone or collectively in response to a variety of guidance cues, including chemical gradients (chemotaxis), adhesion gradients (haptotaxis), mechanical cues (mechanotaxis or durotaxis), and electrical fields (electrotaxis). One paradigmatic example of individual cell movement is the guided migration of primordial germ cells (PGCs) in *Drosophila* and zebrafish. A combination of genetic and imaging approaches showed that the chemokine Cxcl12 directs PGC migration in zebrafish[111,112] and other vertebrates. Led by the chemokine produced by specific groups of cells, PGCs migrate over long distances, while decoy receptors regulate the shape and dynamics of the guidance cue gradient. These studies illustrate the power of combining *in vivo* imaging with genetics to dissect morphogenesis (Figure 3).

In addition to single-cell migration, cells can also migrate collectively in cohesive groups, as in epithelial wound healing or during convergence and extension. Neural crest cells have become a powerful model system to study the mechanisms of collective migration. They migrate and then differentiate into different lineages, including bone, cartilage, and connective tissue. Their morphogenetic mechanisms include chemotaxis, co-attraction, and contact inhibition. In addition, they can be guided by rigidity gradients of the ECM (durotaxis); e.g., *Xenopus* neural crest cells follow a self-generated stiffness gradient in the adjacent placodal tissue.[244] Self-generated gradients can also be of a biochemical nature, where cells can steer themselves by eliminating attractants or repellents in their vicinity (e.g., by endocytosis during lateral line migration).[245,246] In addition, planar cell polarity mechanisms (see below) and supracellular mechanical forces help coordinate the multicellular migratory modes in the lateral line primordium and epithelial tissues. Finally, supracellular constrictions can cause a directed shift of cells together with the surrounding tissue, as during dorsal closure in *Drosophila* embryos.[247] Progress in the field has only been possible through the continuous development of advanced live imaging (Figure 3), specific genetic perturbations (Figure 2), and computational modeling (Box 4) in systems such as *Drosophila*, *Xenopus*, zebrafish, chick, and more recently organoids (Box 1).

A major conceptual advance in the study of morphogenesis has been the view that cell movements can resemble phase transitions in tissue-scale mechanical properties.[154,248] This view has been supported by the development of computational models, such as 3D vertex models (Box 4), and insights from soft/active matter physics. In this scenario, populations of cells can reversibly switch between fluid-like and solid-like behaviors, transitioning from a loosely connected unjammed state that allows rapid cell rearrangements and migration to a jammed state characterized by increased cell-cell adhesion, rigid packing, and limited mobility. In these studies, quantitative mechanical measurements and modeling have been combined to test how tissue-scale properties can influence and coordinate cellular behavior. For example, it has been shown that mesodermal progenitor cells are initially more fluid-like and transition to a jammed state where they will form the future somites.[249] One mechanism by which tissues can be more fluid or stiff depends on adhesion-junctional dynamics, which in turn can be regulated by polarity cues. Such physical contributions add to the increasingly intricate understanding of cooperative biochemical and mechanical mechanisms governing morphogenetic cell migration and will be further described in the section "Tissue mechanics" below.

Further understanding of cell migration mechanisms in morphogenesis remains an active area, enabled by advances in (1) different imaging modalities for real-time *in vivo* imaging with two-photon microscopy and light sheet microscopy; (2) organoids and *in vitro* models that mimic morphogenetic events in controlled environments, which allow force measurements and imaging;[174,250] (3) computational multi-scale models;[251] (4) dynamic *in vivo* manipulation of cells with optogenetics;[169] (5) mechanical perturbations with atomic force microscopy (AFM) and magnetic tweezers; and (6) physical or mechanical measurements of cells and tissues.

Another important component of morphogenesis is the extracellular matrix (ECM). The ECM has long been appreciated as the structural scaffold of tissues and organs. However, the composition and organization of the ECM changes throughout development and influences signaling and morphogenesis. Manipulating ECM components such as fibronectin and laminin revealed their instructive roles during morphogenesis and showed that cell-matrix interactions guide morphogenetic cell behavior events like gastrulation,[65] neural tube formation, and epithelial polarization. Molecularly, key discoveries include the identification of integrins as key ECM receptors and of ECM components as regulators of morphogen movement and signaling. Progress was possible via the use of electron microscopy to reveal ultrastructural details of the ECM like basement membrane (Figure 3) and peptide inhibitor approaches for blocking integrin receptors or matrix components to specifically disrupt cell-ECM adhesions and cell migration. Subsequent efforts revealed the dynamic reciprocity between cells that remodel the ECM and components within it to direct cell function during tissue morphogenesis and homeostasis, e.g., matrix metalloproteinases function in collective cell invasion and neural crest migration. Overall, the ECM was first considered as a cell-adhesive mechanical scaffold but is now viewed as a complex, spatiotemporally active signaling component indispensable in development.[252]

More recent advances provide new insights into the complex and dynamic roles of the ECM in directing embryonic development. For instance, imaging techniques such as second-harmonic imaging of collagens show live changes in matrix properties. RNAi[101] screens discovered that collagen IV is incorporated into the basal membrane, determining organ shape and mechanically constricting cells. The combination of developmental models (Box 1) with strong biophysical approaches also shows how changes in 3D morphology result from differential growth between the epithelial cell layer and its enveloping ECM. At this point, many fundamental questions still remain regarding the instructive role of the ECM and the mechanisms that cells use to sense its composition and mechanical status. For example, do cells have different dynamical ranges at which they sense the ECM mechanical status? Answering these questions will require the development of synthetic tools and imaging technologies combined with force sensors. Moreover, our understanding of the role of the ECM in diseases such as muscular dystrophy (e.g., Duchenne muscular dystrophy) will continue to inform therapeutic approaches.[133–135]

Synthetic approaches have enabled engineering of ECM-like matrices with diverse and tunable biochemical and biophysical properties (Box 3).[91,253] Using such engineered matrices to culture intestinal organoids has shown how dynamic ECM remodeling mechanically directs intestinal morphogenesis. Specifically, ECM remodeling can regulate the activation of the YAP mechanosensing TF and thereby induce reprogramming of organoid cell fates, driving the emergence of distinct intestinal lineages and ultimately tissue architecture.[136] Moreover, *ex vivo* reconstitution assays have shown that reciprocal mechanical interactions between mesenchymal progenitor cells and ECM can create supracellular contractile fluids capable of spontaneous morphological patterning.[254] Overall, these advances and others have revealed that the ECM is not just a structural support for morphogenesis but also functions as an active regulator of developmental signaling, cell fate, and cellular behaviors.[91,92] A major question for future studies is what mechanosensory mechanisms cells use to detect and interpret the spatial and temporal dynamics of the ECM.

We end this section with a discussion of the role of cell and tissue polarity in morphogenesis. Polarization at the single-cell level provides orientation for processes such as oriented cell divisions, coordinated cell migration, and the development of specialized structures such as cilia. Genetic screens (Figure 2) in *C. elegans* identified cortical protein complexes (the Par/aPKC system) that polarize cells, ranging from the anterior-posterior axis of the fertilized egg to the apical-basal axis in epithelial cells.[25,255] The establishment of apical-basal polarity is essential for epithelial tissues and links cell adhesion, cell-matrix interactions, and cytoskeletal remodeling during tissue morphogenesis.[256] For instance, polarization along the apical-basal axis facilitates lumen formation during tubulogenesis of glandular epithelia. Moreover, polarity maintenance involves intricate crosstalk between junction complexes, cytoskeleton elements, and the polarity protein network.

While cell-intrinsic polarity mechanisms can lead to asymmetric cytoskeletal behaviors, they fail to account for long-range coordination of polarity across tissues during development. *Drosophila* genetic screens identified the planar cell polarity (PCP) signaling system, which acts in a cell-non-autonomous manner to synchronize polarity between connected cells along tissue axes.[23,257] The use of mosaic techniques, where the genetic makeup of a group of cells is changed in the context of larger tissues, catalyzed a major shift from considering cells as mere isotropic objects to active units with inherent directionality. PCP signaling guides diverse morphogenetic events including axis elongation, neural tube closure, asymmetric cell division, cochlear hair cell orientation, and ciliogenesis across tissues and organs and underlies a range of birth defects and disease contexts from kidney cysts to deafness.[258] Importantly, PCP is crucial for convergence and extension (CE), as it governs the coordinated orientation and movement of cells within the plane of a tissue.[259] During CE, cells rearrange and intercalate, resulting in the tissue narrowing along one axis (convergence) and simultaneously elongating along a perpendicular axis (extension).[67] PCP signaling ensures that cells move in a directed manner, aligning themselves to achieve proper tissue morphogenesis. This coordinated cell behavior is essential for the development of elongated structures in the body, such as the neural tube and the notochord, and is critical for the overall shaping of the embryo. Molecularly, PCP signaling emerged as a conserved regulatory pathway, including the Frizzled-Dishevelled complex and the Vangl-Prickle complex, that coordinates polarization of cells within the plane of a tissue. A combination of genetic screens and imaging studies identified the key PCP components and discovered their asymmetrical localization between epithelial cell junctions to orient the cytoskeleton. It is now clear that PCP proteins form molecular complexes that communicate orientational information between cells locally.[258] Feedback amplification loops involving these PCP complexes help amplify small orientation biases between neighboring cells into polarized distribution on opposite sides of cells. External directional cues prime the system for robust alignment of polarity across cell fields. The upstream orienting signals that set up the initial bias of the PCP complex can be of both chemical and mechanical nature. For example, actin filaments and microtubule networks become aligned due to mechanical forces, morphogens like Wnts can form tissue-level gradients that orient PCP protein localization directionally, or aligned ECM fibers can provide directional cues. Open questions remain regarding how molecular asymmetries are coordinated among cells and scale up spatially and temporally to establish tissue-level polarity patterns during growth. To address this question, quantitative methods combining single-molecule imaging and fluorescence recovery after photobleaching (FRAP) analysis of dynamics will be essential. Finally, PCP might encompass different processes, as it is still not clear how PCP function compares between planar polarity in epithelial cells and mesenchymal cell behaviors.

TISSUE MECHANICS

The concepts described above are the foundations of morphogenesis and have been complemented over the past 20 years with technologies and concepts that study the physical nature of developmental processes. Physical forces play an instrumental role in directing morphogenesis and help coordinate the morphological changes that shape tissues and organs. While signaling and GRNs are important, physical cues[260,261] or mechanical forces alone can alter cell behaviors and fates.[262–264]

Advances in animal models (Figure 1; Box 1), live imaging, physical measurement techniques, force inference methods, and modeling (Box 4) shed light on the mechanics guiding morphogenesis.[154] Together, these technologies revealed that spatiotemporal patterns of mechanical stress provide precision and directionality to fundamental cell activities, including polarity, migration, proliferation, and differentiation, as tissues and organs take shape. Thus, genetic and biochemical networks may define the molecular toolkit, but physical forces dictate how such tools choreograph embryonic development.[239] Moreover, physical forces can feed back on individual cells regulating the emergence of cellular fates and coordinating cellular behaviors with mechanosensing mechanisms.

Physical forces need to be produced, transduced, received, and interpreted by developing tissues. Actomyosin contractility can generate forces to deform cell and tissue structure, while changes in tissue material properties (e.g., fluidity) might determine the response of cells and tissues to forces.[154] For example, at the beginning of zebrafish morphogenesis, blastoderm spreading depends on a fast and spatially patterned tissue fluidization. This is controlled by mitotic cell rounding and is spatially restricted by local activation of non-canonical Wnt signaling.[265] Environmentally, constraints from surrounding tissues, fluid shear stresses, and other microenvironmental factors create mechanical loads. For example, the developing vertebrate gut tube forms a reproducible looped pattern that is driven by the homogeneous and isotropic forces that arise from the relative growth between the gut tube and the anchoring dorsal mesenteric sheet, which is a tissue that grows at different rates.[266] In addition, forces arising from epithelial growth in 3D confinement are sufficient to drive folding by buckling. Compression can induce cell fate specification in *Drosophila* and other organisms. On a cellular level, the actin cytoskeleton and contractile actomyosin networks provide means to transduce physical inputs into morphological change. One fundamental actomyosin contraction that drives some of the most important tissue remodeling is apical constriction.[267] This allows cells to change their shapes and generate traction that leads to tissue-level morphological changes.

Forces are important also during later stages of development. Postnatally, for example, during crypt formation, apical constriction drives the invagination of intestinal crypts.[268] Deep *in vivo* two-photon-based imaging and complex sample preparation allowed for the visualization of these processes (Figure 3; Box 2). Development of *in vitro* systems for crypt morphogenesis in organoids allowed precise force mechanical measurements, demonstrating how actomyosin mechanical forces are coordinated by osmotic forces and cell migration during crypt morphogenesis.[174,250] Contractile forces generated in one tissue can be received in another tissue and guide morphogenesis of adjacent tissues by creating instructive mechanical microenvironments. For example, actomyosin-driven condensation and stiffening of skin progenitors triggers physical signals prompting fate specification and structural rearrangements in neighboring epithelial cells during the early steps of hair follicle morphogenesis.[269]

In addition to contractile force, the importance of hydrostatic pressure, osmolarity changes, and water re-localization has recently been shown to play a key role in directing morphogenetic movements.[270] Fluid-filled lumens play diverse mechanical and biochemical roles during development, and several experimental techniques have been used to quantify pressure within them. Interestingly, most *in vitro* self-organizing systems have luminal structures that likely support the self-organization of these structures. Generally, lumens form when cells release ions into the extracellular space, generating an osmotic gradient that directs fluid flow. The fluid within the lumen is contained by tight junctions between cells, and its expansion generates hydrostatic pressure that exerts stress on the cells surrounding the lumen or generates isotropic extracellular pressure that deforms the epithelium into buds; e.g., during the development of the otic epithelium.[271]

The mechanisms by which cells sense pressure remain fundamental questions in the field that will require the development of specialized measurement devices for osmolarity and pressure inside cells, lumens, and tissues.[272] One possible sensing mechanism is the deformation of the cell cortex and the plasma membrane by pressure, leading to changes in actomyosin dynamics or activity of ion channels. Other possibilities are changes in cell volume, thus changing the pH, salt content, and concentration of proteins within the cytoplasm.[270] This fluid pressure drives lumen coarsening[273] and cell junction maturation, creating major tissue rearrangements. Pressure also continues to regulate development at later stages of embryogenesis. For example, during development of the lung, pressure drives the formation of complex tissue architecture and influences cell differentiation. Interestingly, cells that line lumens can be subject not only to pressure but also to forces associated with fluid flow.

Flow also generates shear forces: beating cilia generate weak flows, while beating hearts generate much faster flows. During development, shear forces, specifically hemodynamic forces, are necessary and sufficient to induce vessel remodeling in the mouse yolk sac, and forces arising from myocardial contractility and blood flow are essential for heart morphogenesis.[274] In the node, some cilia beat to generate leftward fluid flow, which is sensed by immotile cilia to activate Nodal expression and determine the left-right asymmetry of the body.[98,99,275,276] These discoveries rested on the use of optical tweezers to stimulate cilia and the recording of calcium release in responding cells.

A critical link between mechanical forces and biochemical response is mechanosensation.[91–94] Mechanosensation influences cell behavior and differentiation by allowing cells to detect and respond to mechanical forces. For example, mechanical cues from the cellular environment, such as stiffness, tension, and shear stress, influence the fate of stem cells, directing cell differentiation into specific lineages: soft matrices favor neurogenesis, intermediate stiffness promotes myogenesis, and stiff matrices support osteogenesis.[91] Mechanosensitive structures, like ion channels such as Piezo 2, integrins, and the cytoskeleton, transduce these mechanical signals into biochemical responses. Molecularly, integrins and other mechanoreceptors on the cell surface mediate the transmission of mechanical signals to the cell interior, activating signaling pathways such as Rho/ROCK and YAP/TAZ.[94] For example, in the mouse esophagus, the onset of homeostasis is guided by the build-up of mechanical strain that is mediated by the mechanosensor Yap1 and a switch to differentiation.[277] Another key mechanosensing structure is the nucleus.[278] It can sense forces through connections with the cytoskeleton via the LINC complex, the cytosolic phospholipase A2 (cPLA2), or lamin A/C in the nuclear lamina.

For the future, it will be important to better understand how forces not only shape tissues but also coordinate cellular behavior, providing robustness and directionality to morphogenesis.[279] This interplay or "reciprocal causality"[280] between forces, gene regulation, biochemistry, and potentially metabolism is key in understanding morphogenesis and relies on measuring single cells in full organisms and organoids combined with specific perturbation to address causality.

ORGANOGENESIS

Organs constitute an assembly of cells and tissues that perform specific physiological functions. They are characterized by specialized cells, specific sizes, and complex shapes such as buds, tubes, branches, and networks. Organs form by the complex interplay of numerous developmental mechanisms, ranging from patterning, morphogenesis, and growth to differentiation (histogenesis), homeostasis, and repair. Classic transplantation studies defined tissue interactions that pattern organs, but it was through genetic studies that the molecular foundations of organ formation were uncovered (Figure 2). A major conceptual advance was the discovery of organ selector genes as regulators that dictate the developmental fate of most if not all cells of a specific organ. For example, Pax6 genes are involved in eye development and FoxA genes in foregut development.[18,281,282] Although these genes operate within larger GRNs, they do not simply kick off a cascade of events but rather regulate various aspects of organogenesis over extended time periods.[102] For example, the organ selector pha-4/FoxA regulates not only other TFs but also terminal differentiation genes that create physiological structure and function.[204]

The complex architecture of organs necessitates the coordination and communication between different tissue layers such as mesenchyme and epithelium. This coordination is achieved through a combination of biochemical signals and mechanical interactions. For instance, the branching of the lung is mediated by reciprocal signaling between mesenchyme and epithelium and involves mechanical forces generated by epithelial growth and luminal fluid pressure.[283] Breakthroughs in live imaging have provided exquisite views of organogenesis (Figure 3). For example, it has been observed that heartbeat and blood flow lead to the local activation of mechanotransduction pathways that control the formation of cardiac valves.[274] With improved imaging and optogenetic capabilities (Box 3), it will be possible to both observe and sculpt organ development.

During organogenesis, tissues not only acquire specific identities and shapes, but individual cells develop highly sophisticated structures and functions.[66] This process of differentiation can involve the creation of specialized structures such as cilia, sarcomeres, or vacuoles. However, it also includes the remodeling of shared cellular organelles to fulfill specific needs. For example, the secretory machineries of gland cells (e.g., insulin-secreting cells) and ECM-secreting cells (e.g., cartilage) have to accommodate different cargo and form distinct endoplasmic reticula and Golgi apparatuses.[284] Genetic and genomic studies have suggested that there are two broad regulatory mechanisms underlying differentiation. On one hand, master regulators activate gene modules for specific cellular processes or substructures. For example, devoted TFs activate ciliogenesis or expand the secretory machinery.[203] On the other hand, in a process of phenotypic conversion, different TFs activate the same genes in different cells. For example, not one but several different terminal selector TFs activate genes involved in making neurons glutamatergic.[205]

The advent of large-scale gene expression trajectories can help decipher the gene modules that underlie complex differentiation processes.[206] Genes that are co-expressed and whose proteins interact can be grouped into modules that generate the various aspects of cellular specializations. In the long run, such studies might also inform the creation of cells with newly combined or novel functions. Akin to the transfer of operons between bacteria and the resulting acquisition of new structures and functions, the transfer of gene modules might introduce novel features into cells. More generally, we foresee a renaissance in understanding not only how cells become specified to different fates but also how they differentiate, both *in vivo* and *in vitro*. The close interaction between cell biologists and developmental biologists will catalyze this field of the "developmental biology of the cell."

Differentiated cells in adult organisms face additional challenges compared to embryonic or fetal cells: they have to mature into cells that can adapt to a complex and changing environment.[285] For example, the circadian rhythm of chemical cues contributes to pancreatic islet cell maturation, and the electromechanical stimulation of cardiac muscle contributes to their differentiation. It is still poorly understood how these and other processes (e.g., diet, exercise, oxygen, reproduction, and stress) interface with the GRNs, signaling pathways, and mechanical inputs involved in differentiation. A deeper understanding of maturation processes will require integrating technologies ranging from cell biology to physiology and environmental monitoring. Progress in this field is also essential to generate mature organs that can be used for modeling or treating human disease.

As discussed in "Morphogenesis," proper organogenesis requires communication not just within but also between organs. These interactions go beyond sculpting individual organs and are essential for their proper spatial arrangement in the body. This is most apparent during left-right patterning defects, where organ position can be randomized. For example, applying micro-computed tomography to adult *Drosophila* revealed consistent sex differences in organ shape and position. Sex-specific vascular-like branches control the shape and positioning of the gut, resulting in sexually dimorphic organ geometry.[286] Such "organ origami" provides an exciting entry point to study the roles of organ communication in development and disease.

A major open question in organ formation is the control of growth and size.[287] What determines how quickly an organ grows (growth control) and when it stops growing (size control)? This process is regulated by intrinsic programs that are coordinated with body-wide signals. For example, a pancreas consisting of mouse cells but formed in a rat has the size of a rat pancreas. The intrinsic mechanisms that determine organ size are still poorly understood, but chalones—growth inhibitors—are one way to link organ size to growth. For example, myostatin is generated by muscle to limit muscle growth in a tissue-mass-dependent way.[288] Mechanical feedback has also been proposed to sense organ size, but the exact mechanisms remain unclear.[287] Morphogens have been implicated in organ progenitor growth as promoters of proliferation and inhibitors of cell death, but how (or even if) spatial gradients are sensed during growth control and termination is controversial. In stem cell niches, local exposure to morphogens can restrict growth, and oncogenic mutations can disrupt this process and lead to tumor formation.[289] This process is also observed during cell competition, when cells with lower fitness become "losers" and are eliminated. Such elimination mechanisms can contribute to organ size. In addition, growth termination is also regulated by the ratio of organ size to body size, most likely through a hormonal feedback mechanism that might also be involved in growth precision, which assures that symmetrical structures (e.g., arms or wings) are equally sized.[287] Size control continues to be one of the most mysterious topics in developmental biology and might only be elucidated through the integration of several subfields, including morphogen signaling, metabolism, and environmental influences, and technologies, including measurements of energy consumption, cell growth, and whole-body hormonal signaling.

One of the biggest challenges in the field is the integration of the many factors that influence development into a coherent whole. For example, how is the division, differentiation, or loss of cells coupled to generate the remarkable diversity and reproducibility of tissues[290]? The *C. elegans* invariant lineage tree[42,43] describes in detail how deterministic symmetric and asymmetric divisions, terminal differentiation, and programmed cell death sculpt the organism.[24] Similarly, studies in the *Drosophila* nervous system have revealed how complex structures emerge through a combination of asymmetric divisions, temporal patterning, spatial patterning, combinatorial TFs, biased stochasticity, and apoptosis.[108,291] However, most developmental systems do not have largely invariant lineages, raising the question of how such organs acquire and maintain the correct size and composition.[292] Lineage tracking and clonal tracing studies (Figure 2) have described how differentiated cells are related to each other through division patterns and have revealed multiple mechanisms underlying organ development. For example, in the neural retina, where multipotent progenitors can give rise to several cell types, temporal patterning influences the competence of progenitors to respond to extrinsic signals or creates intrinsic changes that determine cell fate. In addition, stochastic influences, akin to "loaded dice," might create different cell types (e.g., through TF bursts or nuclear positioning). Feedback signals from differentiated cells have also been implicated in regulating cell specification and number. But how these and other elements all combine to generate a structure as organized and complex as the retina remains unclear. Moreover, the development of the retina in different individuals (and even between the left and right eye) will not be identical, raising the question: how many ways are there to make a specific organ, and which mechanisms lead to individuality? The field of "developmental statistics" is emerging to address such questions and has begun to discover specific lineage motifs that are more frequently found than others.

Resolving and combining different models of cell fate acquisition and organogenesis will require the acquisition of near-complete lineage trees and accompanying cellular states (e.g., gene expression, signaling, chromatin, morphology, mechanical forces, and metabolism).[85–87] Improved *in vivo* imaging approaches are beginning to follow cell divisions, reporter gene expression, and mechanical forces. In addition, genome editing approaches are used to introduce cumulative changes into barcodes or endogenous loci to record the lineage relationships between cells (Figure 2). Similar approaches can also be used to record additional biological events, such as the activity of signaling pathways, TFs, or enhancers. Finally, developmental trajectories reconstructed by single-cell genomics can be overlaid with lineage trees to approximate gene expression during fate specification.[185] The combination of such recording and reconstruction efforts is generating global high-resolution maps that form the basis for the field of "integrative developmental biology." With the advent of spatial genomics and metabolomics,[158] these maps are becoming more and more detailed. Beyond technological challenges, such as the construction of lineage trees and the recording of biological events, these approaches require complementary computational, theoretical, and modeling efforts. At the practical level, how can we reconstruct complex lineage trees, especially if sampling is sparse? How can we reconstruct gene expression trajectories that reflect *in vivo* transitions? Many powerful approaches have emerged in recent years, but how to best deal with multidimensional datasets is still contentious.[293,294] At the conceptual level, how should the transition of cells and tissues during development be portrayed? The dominant visual metaphor is still the Waddington landscape of hills and valleys[295] through which cells travel like rolling balls. Resembling Waddington's landscape, current depictions of single-cell trajectories often use state manifolds that attempt to represent in high-dimensional space the possible states a cell can occupy and the trajectories it can follow. How well do these and other depictions capture the underlying biology? How can these representations lead to specific hypotheses and predict the effects of perturbations? How can they be the foundation for general concepts? While many useful and interesting observations have emerged from these efforts, it continues to be an exciting challenge for the field to move from catalogs, atlases, and landscapes to novel concepts and theories.[296]

STEM CELLS, REPAIR, AND REGENERATION

Once formed, organs are maintained through a dynamic balance between cell proliferation, differentiation, and elimination.[290] While the field of developmental biology has often focused on embryos (or imaginal discs) as model tissues, recent decades have witnessed an extension toward the analysis of adult tissue development (Figure 1; Box 1). Cellular turnover, regulated by stem cells and their niches, is a key aspect of this process. Since the pioneering studies in the hematopoietic system and skin, the combination of genetics, clonal tracing, transplantation, and imaging has led to the identification of various stem cells that generate and maintain tissue through cell division.[44–46,297] While in some systems (e.g., *Drosophila* neuroblasts), each stem cell divides asymmetrically to create one daughter that remains a stem cell and another daughter that proliferates and/or differentiates, in many other systems, stem cell maintenance and differentiation is mediated at the population level. In these scenarios, stem cell division can lead to initially equipotent daughter cells whose fate is regulated in part by space constraints, access to a niche[47,48] that actively maintains the stem cell fate, and competition for niche-derived signals. Labeling studies have revealed that this process can result in neutral drift, where some stem cell clones are lost while others expand. For example, multicolor pulse labeling of stem cells in the intestine resulted in the clonal dominance of a few stem cells.[149] Similarly, the generation of multiple cell types from stem cells is rarely controlled by a strict lineage hierarchy; it is a highly plastic process that is influenced by intrinsic differences between stem cells and the diverse environments they populate.[298] The emerging picture suggests that organogenesis and homeostasis rarely follow a deterministic tree-like process. Instead, they resemble a landscape characterized by heterogeneity and plasticity and regulated by local cues and feedback mechanisms that control the proportions, locations, and elimination of stem cell descendants. Despite recurrent lineage motifs, most lineages are not determined but are probabilistic and statistical. Recent progress in the DNA-based recording of lineage relationships (Figure 2) will help reveal the developmental statistics of cellular specification during organ formation and maintenance.

Another hallmark of organ homeostasis is the remarkable plasticity of cells and the ability to repair damaged tissue. For example, the endometrium sheds during menstruation but is rebuilt rapidly without scarring, and the injured intestine can repair its epithelium in a few days.[289] How is such repair controlled? In some systems, resident stem cells constantly replenish the different cell types needed for repair (e.g., hematopoietic stem cells [HSCs] and satellite cells). In other organs, differentiated cells can dedifferentiate to fetal-like progenitors and then give rise to the major cell types needed during repair (e.g., intestine). In some cases, differentiated cells can themselves divide and replace lost cells (e.g., hepatocytes) or dedifferentiate to become stem cells (e.g., bone).[290] These repair processes are often regulated by neighboring cells. For example, immune cells such as tissue-resident macrophages can activate quiescent progenitors (satellite cells) at repair sites and promote stem cell division and differentiation. Stem cells themselves can also be subject to repair mechanisms. During zebrafish hematopoiesis, macrophages can interact with stem cells, remove cytoplasm, and promote survival.[299] *In vivo* imaging approaches and lineage tracing approaches in injury models will continue to provide deep insights into such repair processes.

Repair processes are also linked to tissue remodeling during physiological and disease processes. For example, hypoxia results in increased erythrocyte production, stress can induce hair graying by depletion of melanocyte stem cells, modulating glp-1 signaling can change adipose tissue (and has become a revolutionary drug treatment), and mating can remodel the digestive system. In turn, environmental insults can overtax repair processes and promote inflammation and tumorigenesis,[300] but notably, somatic mutations can also be protective and promote regeneration, e.g., in liver disease. The interface between tissue repair, endogenous physiology, and environmental influences is one of the most complex fields in developmental biology. Future progress requires technologies that scale from molecules and cells (e.g., detection of damage to DNA or cells induced by environmental pollutants) to tissues and whole organisms (e.g., local interactions in stem cell niches, inter-organ crosstalk, and body-wide immune responses).

Extreme cases of regeneration—from organs to appendages—involve the repatterning of major tissue portions, such as for zebrafish heart regeneration or axolotl limb regeneration.[34–38,115,116,297,301–303] Such repatterning programs often use the same or similar genes that were used during embryonic patterning. For example, after heart injury, adult cardiomyocytes can dedifferentiate to acquire a near-embryonic state and temper their contractile machinery.[37,38] However, before reactivating embryonic programs and following initial damage, cells go through an adult-specific regeneration phase that often includes wound repair, proliferation, and blastema (a regenerative outgrowth at wound sites) formation to create a pro-regenerative environment. To help initiate regeneration programs, genomics studies have identified regeneration-specific enhancer elements that drive the expression of factors involved in regeneration. These enhancers can direct expression locally to the regenerating organ or cell type, whereas others can control gene expression in organs distant from the injury. Such inter-organ communication allows for body-wide coordination of regeneration (and is also involved in coordinating the development of different organs). For example, if a structure is injured during development, other tissues might slow down their growth or differentiation to let the regenerating organ catch up. In other cases, the secondary response might feedback onto the primary site of regeneration. Such remote regulation is often mediated by hormones (e.g., Dilp8)[287] or soluble factors (e.g., vitamin D), but many of these relay signals remain to be discovered.

Even more dramatic than organ regeneration, some species such as Hydra and planaria can regenerate a whole organism from small body parts. In planaria, this process relies on neoblasts—stem cells that can give rise to dozens of different cell fates—and a prepattern mostly represented by pre-existing muscle tissue.[117,302] Akin to the coordinate systems set up in embryonic tissues by signaling molecules, adult planaria harbor positional patterns along the body axes. After tissue injury or ablation, the coordinate system is restored, involving wound signaling and neoblast proliferation, and cell types are restored through neoblast specification. In a subsequent process of cell migration, sorting, and differentiation, organs are reformed.

The ability of organs and organisms to repair and regenerate varies widely among different species and organ systems. In humans, organs like the liver and skin have a high regenerative capacity, whereas others, such as the adult heart and central nervous system, have limited regenerative abilities. It remains in many ways a mystery as to what determines regenerative ability. Comparative evolutionary developmental (evo-devo) studies in different model systems (Figure 1; Box 1) combined with large-scale genomics and genetics (Figure 2) might provide entry points to answer this question.[304] For example, emerging model

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systems such as spiny mice have a remarkable ability to heal large skin wounds without scarring.[305] The potential conservation or absence of GRNs for wound healing and repatterning in regenerating and non-regenerating systems, respectively, might reveal ways to confer or improve regenerative abilities to non-regenerative organs and organisms. Which systems are best suited for such comparative studies remains to be identified.[306]

Although often not considered a part of normal development, aging and death are the final phase of organ development and homeostasis. Aging is a failure of homeostasis and repair, with hallmarks that include accumulation of mutations, reduction of stem cell number and proliferation, fibrotic ECM, senescence, and changes in metabolic function.[307] Genetic studies in *C. elegans* identified genes involved in aging and lifespan determination,[78–80] connected these aging genes to environmental cues (e.g., diet), and raised the possibility that lifespan can be prolonged. However, it is still poorly understood how some cells and organisms can delay aging or how germ cells can live forever and upon fertilization form a zygote that is young (again).[97] One intriguing approach toward rejuvenation is the activation of developmental programs or the exposure to factors active in younger animals. The study of parabiotic mice[118] and the expression of pluripotency factors[107,308] have identified putative rejuvenation factors, but in a background of accumulated mutations in older cells, such interventions might increase the rate of malignant transformations. Despite these concerns, it is conceivable that development of an entire organism can be slowed or even reversed through a combination of rejuvenating factors, the removal of senescent cells, the elimination of damaged organelles, or the increased maintenance of genome integrity. Emerging model systems that age rapidly (killifish) or slowly (naked mole rat) might help define anti-aging mechanisms (Figure 1; Box 1).

There is currently much excitement and investment in the possibility of rejuvenating humans. Given the many factors and organ systems involved in aging and the already long lifespan of humans, the field will face considerable challenges in providing unequivocal evidence that a specific rejuvenation treatment is safe, specific, and desirable. Who will be able to afford or volunteer to test treatments that might cause cancer or rejuvenate one organ (e.g., the heart) while other organs (e.g., the brain) keep declining? This field will provide fascinating insights into repair processes and homeostasis, but direct applications in humans might not be immediate.

Arguably, the study of stem cells has had some of the most direct impact on developmental biology in understanding and treating human disease. Stem cell biology has revealed mechanisms of carcinogenesis[309] and has led to approaches for tissue replacement and repair. For example, HSC transplantation followed by *in vivo* expansion is routinely used to restore the blood system after chemotherapy or radiation therapy; epithelial stem cells are employed *in vitro* to generate skin suitable for transplantation onto burn victims; and salivary gland stem-cell-derived organoids are being transplanted into cancer patients following irradiation.[32,33,289] Similarly, the generation of embryonic[51,52] and induced pluripotent stem cells (ESCs and iPSCs)[107] has provided a starting point to generate specific cell types *in vitro* for disease modeling and cell therapy.[109,110] The current convergence of genomics (Figure 2), imaging (Figure 3), and engineering approaches (Figure 4) has the potential to build more robust *in vitro* models to study human disease and create mature organs for transplantation into patients.

SELF-ORGANIZATION

In addition to the many roles described above, stem cells have a strong self-organization capacity, illustrated by their role in forming *in vitro* embryonic models and organoids (Figure 1; Box 1). In general, all cells, from bacteria to mammals, can self-organize, pattern, and arrange themselves into higher-order structures by local interactions and tissue-level feedback. This process plays a fundamental role in development and lies at the heart of the coordination, robustness, and adaptability found in cellular and organismal organization.[280] Self-organization refers to the emergence of overall order in a given system as a result of the interactions and coordination of its individual components. This concept has been recognized as a core principle in pattern formation in multi-component systems of the physical, chemical, and biological world.

One problem in defining the exact contribution of self-organization in developmental systems stems from the rigorous prerequisite that self-organization develops without prepatterns and external cues. Therefore, it has been experimentally challenging to segregate the pure self-organization potential of cells from the extrinsic control by morphogens, mechanical forces, and other inputs. Indeed, self-organization, prepatterns, and extrinsic controls co-exist *in vivo*. The development of 3D self-organized organoids has demonstrated that homogeneous populations of cells—even starting from a single cell—have the intrinsic ability to generate emergent, self-organized behavior resulting in the formation of complex multicellular asymmetric structures.[81,82,120] Since the first organoids[82,310] the field has been developing many *in vitro* systems to follow self-organized tissue organization in a controlled way,[140,311] both from adult stem cells recapitulating regenerative response or from embryonic stem cells modeling developmental pathways (Box 1). Stem-cell-derived systems can self-organize into embryo-like structures without maternal contributions or extra-embryonic tissues[143,144,312]; other organoids are first formed by direct differentiation of the different germ layers in 2D or 3D embryonic bodies to then self-organize into their definitive tissues. In a dramatic example of self-organization, a single adult intestinal stem cell can generate complex intestinal organoids with the cell type composition and spatial organization of the tissue of origin.[82] This process relies on the plasticity of cells and the expression of the mechanosensor YAP1. A regenerative response drives the re-emergence of all cell lineages by sensing tissue-scale mechanical properties.[136] Self-organization then continues when the newly generated cell types create a differential curvature of the tissue due to the relocalization of the actomyosin network. A self-induced, tissue-intrinsic mechano-chemical feedback then drives the emergence of the crypt.[174] These examples illustrate the remarkable propensity of some cells to self-organize.

One important aspect that is apparent in self-organizing systems is the role of non-genetic cell-to-cell variability. This variability generates heterogeneous cellular states in populations of cells and tissues. In turn, these cellular states sense external cues and respond differently to the same environment, resulting in asymmetries in cell differentiation and morphogenesis.[136] These observations raise the questions of how cell-to-cell variability in cellular states contributes to self-organization and how these states lead to different responses to external cues.[170,313] To answer these questions, the field will require single-cell methods that go beyond single-cell transcriptomics, as differences in cellular states are most apparent at the level of protein expression and localization and might depend on intracellular organelle composition as well as metabolic and signaling states.[314] Thus, spatial proteomics and metabolomics[315] at single-cell resolution, lineage tracing, and high-content multiplexed imaging technologies at the population level will play an important role in the study of self-organization.

As discussed above, self-organization *in vivo* is difficult to disentangle from other mechanisms; however, some experimental approaches have been able to reveal self-organization during morphogenesis by combining mechanical measurements and specific perturbations. For example, increased luminal pressure in the mouse blastocyst stretches the cells of the lumen, which increases the cells' cortical tension. In turn, the increased cortical tension induces tight-junction maturation, which allows the cells to accommodate the lumenal pressure buildup.[316] Lumen formation also contributes to the self-organization of the lateral line in zebrafish. As a result of the interplay between lumen formation and FGF signaling, the migrating lateral line primordium deposits rosette-like organs at its rear. These examples illustrate how lumen formation can drive self-organization.

In vivo self-organization is not only found within the same tissue but also between neighboring tissues. For example, during avian development, the skin transitions from a uniform tissue bilayer to a bilayer covered with multicellular follicles. This process is driven by local contractility and the spontaneous aggregation of mesenchymal cells. Aggregation then locally compresses the adjacent epidermis and generates the shape and force to activate expression of follicle-specific genes.[269] This example shows how self-generated mechanical forces can contribute to the self-organization of tissues.

Another emerging model system for the study of apparent self-organization are annual killifish. Remarkably, during early embryogenesis, blastomeres de-adhere and disperse and only aggregate later to form an embryonic axis. Recent studies suggest that these fish lack the prepatterns provided by maternal gene products found in other fishes and self-organize as blastomeres aggregate.[317] This process is reminiscent of the seemingly spontaneous axis formation observed in embryo-like models *in vitro*.

The study of self-organization is still in its infancy and will require several technological improvements. High-resolution live imaging and reporter genes are necessary to follow entire tissues as they self-organize (Figure 3; Box 2). Perturbations that specifically disrupt biochemical versus mechanical factors are needed to determine what forces drive self-organization. Synthetic approaches (Figure 4; Box 3) are required to reduce systems to their minimal components. The back-and-forth between *in vivo* and *in vitro* models will reveal shared and divergent mechanisms of pattern formation and morphogenesis.

METABOLISM IN DEVELOPMENT

Metabolism provides the energy and building blocks needed for cell growth, division, and differentiation.[318] In the field of animal development, studies of metabolism have seen remarkable progress in the past years. First, it has become clear that metabolic shifts meet changing bioenergetic and biosynthetic demands during development. Second, metabolism does not simply have permissive housekeeping functions but can also regulate developmental states and transitions. In the following section, we provide examples of various metabolic roles and the technological challenges the field is beginning to overcome.

Metabolic processes have been analyzed through direct measurement of metabolites, isotope tracing, or extracellular flux assays. Such studies revealed the transition between glycolysis and oxidative phosphorylation to be one of the most widely observed metabolic switches. While differentiated cells often use oxidative phosphorylation, many proliferating cells, such as embryonic cells and progenitor cells, display high aerobic glycolysis. This process was first described as the Warburg effect in cancer cells but is a general feature of many rapidly dividing cells. Despite the broad use of Warburg metabolism, it is unclear why cells use a process that produces much less ATP than oxidative phosphorylation and generates high levels of lactate.[319] It is possible that glycolysis meets the demand for metabolites other than ATP, such as NAD+, and for the synthesis of phospholipids, amino acids, and nucleotides to generate new biomass via glycolysis shunts. It has also been proposed that lactate might serve as a driver of the cell cycle[320] or might be shuttled as an energy source to neighboring cells.[321] Conversely, glycolysis might be used instead of oxidative phosphorylation to prevent the generation of potentially toxic byproducts, such as radical oxygen species. In this latter strategy, the use of specific metabolic pathways is not only to support the metabolic demands of a cell but also to avoid unwanted side effects such as DNA damage in stem cells.

While the Warburg effect is associated with proliferation, the reduction of metabolic features that drive proliferation is associated with quiescence, when cells have entered a state of reversible cell-cycle arrest. These cells display reduced metabolic flux, low mitochondrial activity, and decreased protein production, and manipulation of these processes can result in exit of quiescence.[322] Thus, different metabolic states are associated with distinct developmental states, and metabolic switches mark developmental transitions. As the detection of metabolites and metabolic flux becomes more sensitive, we foresee the systematic generation of high-resolution metabolome atlases. Such atlases will be the basis for defining shared and divergent metabolic features and for testing the effects of metabolic disruptions. In addition, future studies will need to distinguish between bioenergetic and biosynthetic roles. Such a thermodynamic or energy-expenditure view of development (e.g., ATP consumption) can be gained by calorimetric measurements, oxygen consumption rates, extracellular flux assays, or pyruvate fluorescence resonance energy transfer (FRET)-sensor and NAD+/NADH ratios with imaging metabolic probes.[157,323]

Metabolism is not only required as a source of bioenergetic and biosynthetic needs but also plays regulatory roles.[324,325]

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Such regulation can be global by affecting cellular functions such as chromatin modifications, or it can be specific by regulating small sets of proteins or signaling pathways. As an example for the former concept, histone acetylation, a modification associated with increased chromatin accessibility, is regulated by the metabolite acetyl-CoA, which serves as an acetyl donor. Thus, increasing acetyl-CoA can maintain embryonic stem cells in a pluripotent state, whereas decreasing levels are often associated with differentiation.[326] In another example, the TCA cycle metabolite alpha-ketoglutarate can serve as a co-substrate for dioxygenases that catalyze the demethylation of methylated histones or DNA. Thus, alpha-ketoglutarate can promote the differentiation of various cell types, including epidermis and intestine.

These examples illustrate how specific metabolites can modify global epigenetic processes during development. It is unclear, however, if metabolism is permissive or instructive in eliciting developmental transitions. Are metabolic changes simply one of many downstream effects of more traditional regulatory steps, such as signaling pathways or TFs, or do metabolic changes provide a powerful strategy to coordinate global changes in developmental states? Investigations of metabolic reprogramming during tumorigenesis have provided examples of how to address such questions. For example, a combination of metabolic profiling and functional studies revealed that high levels of arginine promote tumor formation by regulating a specific factor (RBM39) that controls the expression of metabolic genes.[327] These studies illustrate how specific metabolites can act instructively.

Metabolic switches can directly regulate specific signaling pathways. For example, in the presomitic mesoderm, the interplay between developmental signaling pathways and metabolic states can guide differentiation.[328] This system displays a posterior-to-anterior gradient of FGF signaling, Wnt signaling, and glycolytic activity. In a positive feedback loop, FGF increases glycolytic activity, which in turn increases Wnt signaling, which increases FGF signaling. This process promotes mesoderm formation and tail elongation and has also been implicated in regulating the speed of the somite segmentation clock.[329] In another example, different sphingolipids are associated with different dermal fibroblast states.[315] This lipotype heterogeneity modulates cell identity by modulating FGF signaling, wherein some sphingolipids promote and others repress FGF signaling. In turn, FGF signaling promotes the production of sphingolipids that activate FGF signaling. Thus, a positive feedback loop helps generate distinct fibroblast types.

These studies provide examples of the crosstalk between signaling and metabolism during development and illustrate how technological advances open new frontiers. For example, metabolism can be coupled to cell state by combining single-cell transcriptomics with high-resolution matrix-assisted laser desorption/ionization mass spectrometry imaging (MALDI-MSI).[315] In addition, fluorescently labeled probes, such as bacterial toxins, can be used to detect specific classes of metabolites, and improved sensors for specific metabolites can be used in live cell imaging (Box 3). In the future, the field will routinely employ assays to measure metabolic reactions through imaging and isotope tracing. For example, recent studies have pioneered new approaches to perform isotope labeling in the placenta and embryo during mid-gestation and mid-to-late gestation in mice, opening the possibility of monitoring embryonic metabolism.

Another important challenge is the analysis of metabolomic heterogeneity to discover metabolic interactions between different cells and across organs.[330] For example, in the brain, glucose metabolism is compartmentalized between neurons and astrocytes. Neurons show a lower glycolytic activity than astrocytes. This compartmentalization generates a gradient of the glycolytic end-product lactate from astrocytes to neurons. This lactate shuttle allows neurons to use glucose preferentially for the maintenance of cellular redox balance, rather than for energy production. An analogous metabolic interplay has been observed in the intestinal organoids with the shuttling of lactate from Paneth cells to intestinal stem cells to regulate mitochondrial respiration and reactive oxygen species (ROS) production to drive crypt formation.[321] The mechanistic understanding of this cooperativity will require techniques such as spatial transcriptomics and metabolomics to profile metabolic diversity in complex tissues.[158] Finally, it will be necessary to develop methods to induce single-cell perturbations in complex multicellular systems to address whether changing metabolic flux in one cell type impacts the behavior of the rest of the tissue. Such functional assays are complicated by the many roles a given metabolite or metabolic enzyme might have. For example, many metabolic enzymes, especially all glycolytic ones, display "moonlighting" functions (e.g., non-enzymatic nuclear roles), and many glycolytic enzymes bind to RNA. Some of these proteins fulfill nuclear functions that are distinct from their roles in glycolysis dependent on their localization in the cell. Thus, a major challenge in the field of developmental metabolism will be to distinguish between roles in bioenergy generation, biosynthesis, and other functions depending on the localization of the protein, not only in specific cell types but also in specific cellular compartments. This will require specificity in perturbations to add mechanistic insight into causality in metabolic fluxes, enabling more direct evidence for instructive functions of metabolism. To assess the specificity of these perturbations, it will be essential to combine them with visualization approaches to measure metabolites and metabolic fluxes in living cells within organoids and model systems.

DEVELOPMENTAL TIMING

The processes we discussed above unfold in ordered fashion over time and can take seconds (morphogen transduction), days (embryogenesis), months (maturation), or years (aging). However, seemingly similar processes can take orders of magnitude longer in some species than others (e.g., the time from fertilization to fertility). How is such developmental timing controlled?

There are two major aspects to developmental timing: temporally ordered transitions (e.g., specification before differentiation) and the tempo or duration of development (e.g., fast in *C. elegans* but slow in humans). Classic molecular genetic studies have shown that order is often generated by cascades of regulatory steps, wherein A regulates B, which in turn regulates C. Such temporal order can be used to generate cellular diversity and often becomes irreversible through changes in chromatin (and other processes). For example, in the developing nervous system, the consecutive activation and repression of temporal identity factors determines neuronal fate.[331] During developmental transitions, hormones or microRNAs can regulate large cohorts of genes and thus drive and coordinate temporally ordered changes. For example, during the maternal-to-zygotic transition, a class of microRNAs represses and eliminates hundreds of maternal mRNAs and helps the embryo to transition from development driven by a pool of maternal gene products to embryogenesis regulated by zygotic RNAs. How exactly such global temporal order unfolds is still poorly understood, but high-resolution genome-wide approaches are beginning to reveal intricate regulatory interactions.

Several mechanisms that control the tempo of events have been discovered.[332] At the global level, species develop at very different rates. *In vitro* culture as well as inter-species transplantation experiments suggest that this tempo is often cell-intrinsic. Manipulations of metabolic rates (e.g., NAD+/NADH ratios) can slow down or accelerate development. For example, the speed of the segmentation clock[95,96]—an oscillatory system of gene expression that is translated into spatial periodicity of somites—depends on NAD+ redox balance and downstream translation rate.[333] Moreover, NAD+ redox homeostasis seems more important than ATP availability for regulating growth tempos and oscillation rates. Increasing ambient temperature or reducing glycolytic flux can also speed up the clock. Moreover, the pace of neuronal development depends on the rate of metabolic activity of the TCA cycle and the oxidative activity in mitochondria.[334]

Metabolic rate is likely a major source of temporal diversity and ideally suited for the regulation of developmental speed. It affects most if not all biochemical reactions but can be locally adjusted to affect tempo in different regions, tissues, or cell types. A subroutine of the metabolic rate mechanism is the adjustment of protein synthesis and degradation to influence tempo. For example, *Drosophila* minute mutants, which affect ribosomal proteins, or *C. elegans* rDNA mutants grow more slowly. These studies show how modulating the activity of "housekeeping" functions can modulate developmental tempo. Although glycolysis has been linked to the activity of morphogen signaling pathways such as Wnt, how metabolic rate can become an instructive mechanism to regulate tempo is largely unclear. As discussed above, such studies would require perturbation technologies that decouple energetics from signaling.

Another large-scale mechanism for temporal control involves chromatin modifications. For example, flowering in plants is only initiated after long-term cold exposure, a process called vernalization.[335] During winter, a Polycomb-based switching mechanism represses the expression of an inhibitor of flowering by promoting slow chromatin closing. Thus, long-term cold exposure de-represses an activator of flowering. Repressive chromatin has also been implicated in the timing of maturation of human cortical neurons, where transient inhibition of repressive chromatin can accelerate maturation *in vitro*.[336] Analogous processes of chromatin opening and folding contribute to the temporal control of multiple genes located *in cis*, such as Hox or globin genes. In these systems, genes are activated consecutively based on their respective locations in the genome, resulting in the translation of genomic location into temporal information.

There are many other fascinating phenomena related to developmental timing: bamboo can flower at intervals of up to 120 years; cicadas emerge every 13 or 17 years; pregnancy leads to the timed generation of neurons and changes the behavior of the mother[337]; and many animals can enter long periods of arrest or diapause. Our understanding of the molecular nature of these and most other developmental timers is unknown but will be greatly helped by long-term imaging studies, genomics, and *in vitro* systems.

Changes in developmental timing have been implicated in the evolution of morphological differences. In this process of heterochrony, extending or reducing the growth period of a tissue can change its size and shape. For example, accelerated or delayed development of different jaw regions is thought to underlie the diversification of jaw structures and feeding mechanisms in cichlids.[338] The application of genomics and genetics approaches (Figure 2) to different model systems (Figure 1; Box 1) will help reveal the mechanisms of heterochrony in development.

ECOLOGICAL DEVELOPMENTAL BIOLOGY

The genome contains the blueprint for the development of organism-specific features. While the environment (e.g., nutrients, temperature, etc.) is often thought to have a mostly permissive role in interpreting this blueprint, ecological factors can instruct the phenotype in systems that are developmentally plastic. For example, plant cells contain the same genome, but in soil, these cells will form roots; in the immune system, exposure to environmental pathogens or pollutants can induce complex differentiation programs; in the nervous system, external cues and enriched environments influence development during critical periods and induce cellular changes to form memories; in the digestive system, nutrient availability or dearth can remodel tissues; whether a female bee or ant becomes a fertile queen or a sterile worker depends upon the diet that the larvae receives; and the size of a male dung beetle's horn depends on the quality and quantity of the dung that the larvae ingests. The field of ecological developmental biology studies the developmental consequences of organisms interacting with their environment, including other organisms.[339] In this section, we provide some examples of organism-environment interactions during development.

Temperature is one of the most fundamental environmental influences on development.[340] The temperature dependence of biochemical reactions accounts for the often-observed acceleration of development when an organism is exposed to higher temperatures. This phenomenon is most apparent within the same embryo when microfluidics is used to generate different temperatures along the body axes: regions exposed to higher temperatures develop more rapidly than regions of lower temperatures.[341] However, at the same temperature, different organisms can have different developmental rates, revealing the presence of species-specific factors that regulate developmental tempo (see "Developmental timing"). Temperature can also be an environmental stressor that affects the entire developing system or has cell-type-specific effects. For example, the developing zebrafish notochord is more sensitive to temperature increases than other tissues, most likely due to higher secretory activity and reliance on the unfolded protein response compared to other cell types.[340] Robustness to higher temperatures can also be conferred by seemingly redundant enhancers, which at normal temperatures can be deleted individually but at elevated temperatures are essential.[200]

Temperature can also be used to generate phenotypic diversity. A particularly striking example of temperature-dependent development is the gaudy commodore butterfly. They have such strikingly different pigmentation patterns in the winter and summer that they were assumed to be two different species. Temperature also influences sex determination in reptiles and fishes. For example, many sea turtle species tend to produce more male offspring at lower temperature, an effect that appears to be mediated through a calcium-Stat3-chromatin modification axis.[342] Thus, the study of temperature has and will provide interesting insights into the role of environmental factors in developmental plasticity, resilience, and robustness.

Environmentally adverse conditions can suspend or modify development. For example, diapause is a physiological state of arrested development that allows organisms to survive adverse environmental conditions, such as extreme temperatures, lack of nutrients, or anoxia.[343] The regulation of diapause is best understood in *C. elegans*, where high population density, limited food supply, and high temperatures induce complex hormonal and molecular pathways to arrest larval development. Killifish exhibit a fascinating form of diapause that is an adaptation to aquatic habitats that can dry up completely. The diapause in killifish embryos allows them to survive periods of drought for several months or even years. The adults die, but the diapause embryos survive until the ponds fill up again. Several mammalian species, including some marsupials and rodents, also undergo diapause, which involves the suspension of embryogenesis at the blastocyst stage before implantation. The diapause allows the young to be born during the spring, when food is plentiful.

Some animals can adapt to environmentally adverse conditions by remodeling their organs. For example, upon feeding, snakes undergo a rapid and substantial increase in the size of their digestive organs, whereas these organs atrophy during fasting. In mice, cold exposure results in growth of the intestine.[344] In preparation for long flights during migration, some birds reduce their digestive system to reduce weight. Environmental cues can even be communicated between generations to enhance developmental robustness. For example, *C. elegans* can sense an adverse environment and adjust maternal provisioning—the transfer of cytoplasm—to promote normal development of their offspring. However, adverse conditions can also lead to maladaptive developmental responses. For example, air pollution can change lung homeostasis through direct damage and inflammation, which in turn can lead to abnormal repair and fibrosis as well as tumor formation.[300] In a world of changing ecosystems, studies of normal and pathological developmental responses to environmental challenges is becoming an important branch of developmental biology.

A particularly fascinating area of ecological developmental biology is the phenomenon of mutualistic symbiosis—the beneficial close association of different organisms.[345] Famously, the interaction between plants and their microsymbionts leads to morphological changes that generate nodules for nitrogen reduction and assimilation and metabolite exchange; interaction with the bacterium *Vibrio fischeri* induces the formation of the light organ in bobtail squid; and microbial symbionts modulate the maturation of the mammalian intestine, immune system, and nervous system. The use of new model systems and genomics approaches will help uncover many more examples of such symbiotic relationships and might help employ symbionts and other environmental factors to optimize tissue differentiation and repair.

Evolutionarily, environmental factors play important roles even in the most simple developmental systems and might have been initial triggers for multicellularity and cell type diversification. For example, in choanoflagellates—the closest living relatives of animals—environmental cues (generated by bacteria) can trigger colony formation (a potential precursor to multicellularity), diversify cell types (a potential precursor of differentiation), and even induce morphogenetic movements (a potential precursor to gastrulation). In more complex systems, evolution appears to have selected for self-sufficient development during embryogenesis by providing yolk-rich eggs or a placenta. In contrast, differentiation, maturation, repair, and remodeling reveal the developmental plasticity that responds to environmental cues. Symbionts that help generate the rumen and then digest plant material within it have facilitated the evolution of obligate mammalian herbivory. A fascinating area of future research is the understanding of the ways ecology sculpts the evolution of morphological diversity.[346] Given the complexity of natural ecosystems, one potential strategy to simplify future studies is to reconstruct reduced ecosystems in the lab.[347]

The examples above reveal strong links between developmental timing, metabolism, and environment. Thus, in the era of anthropogenic climate change, the field of ecological developmental biology will become increasingly important in areas such as agriculture, aquaculture, and ecosystem maintenance. The developmental changes that transform solitary grasshoppers into gregarious plague locusts are mediated by climate changes, and changes in temperatures could cause extinctions through sex determination schemes or phenology. The current technological breakthroughs in genomics and imaging will provide a more systematic understanding of development in different environments, and progress in engineering and synthetic biology has the potential to create features that might allow development in extreme terrestrial (or even extraterrestrial) environments.

EVOLUTIONARY DEVELOPMENTAL BIOLOGY

The processes described above evolved over hundreds of millions of years. Evolutionary developmental biology aims to understand how morphological similarity and diversity evolved. By comparing different organisms using genomics, genetics, and imaging, evo-devo can reconstruct ancestral relationships, identify conserved mechanisms, and discover how new phenotypes emerge. Moreover, comparative approaches can help uncover fundamental concepts and mechanisms. One of the major insights in the past 50 years has been the remarkable evolutionary conservation of developmental mechanisms. For example, lateral inhibition and progenitor maintenance are often mediated by Notch signaling; symmetry breaking and the polarization of tissues is often associated with Wnt signaling; patterning along the anterior-posterior axis often employs the Hox clusters; and master regulators of cell types and organ identity are conserved. The discovery of such broadly shared toolkits suggests that there are fundamental solutions to challenges in the development of an organism. We already discussed the diversification of body plans and cell types through regulatory changes.[114,208] Here, we discuss efforts that aim to reconstruct evolutionary novelty—the appearance of new genes, new cell types, and new structures.[348]

At the gene level, large-scale sequencing efforts have shown that novel genes can arise through multiple different mechanisms, including the neofunctionalization of duplicated copies, the rapid sequence evolution from non-coding sequences, or the co-option of viral or transposable elements.[349] This genomic novelty often correlates with the emergence of novel structures and cell types. For example, clade-specific genes in cnidarians are associated with the development of stinging cells, and the emergence of Myelin Basic Protein is linked to the emergence of myelin sheaths. One potential challenge in this field is the determination if a particular gene is "new." To address this issue, gene and genome comparisons can now be combined with large-scale structural comparisons.[199]

At the level of cell types and organs, the cooperation and interdependence of diverse cell types is the foundation for organ evolution. Diverse cell types might evolve from multifunctional "ancestral" cells that diversify into "sister" cells with different subfunctions. In this model, akin to gene duplication followed by subfunctionalization, functional segregation or neofunctionalization of cells can underlie cell type diversification.[350] For example, a chemosensory-neurosecretory precursor is likely to have existed in ancestral chordates before the diversification into separate chemosensory and neurosecretory cells in vertebrates.[351] This "subfunctionalization model" is complemented by the "niche creation model," in which one cell type forms the foundation for the evolution of a second cell type.[352] For example, in rove beetles, the defense gland consists of two cell types that cooperate to synthesize and release irritants. It has been proposed that an initial cell type created a weak defense system and was then complemented by a second cell type that contained a stronger irritant but whose solubility was dependent on the initial cell type.[352]

Evolution has not only diversified genes and cells but also morphologies. Strikingly, morphologies and morphogenesis can differ remarkably despite similar cell types, GRNs, and patterning mechanisms. For example, gastrulation morphologies—the topology of germ layers as mesendoderm progenitors internalize and the main body axes emerge—can differ widely between organisms.[353] Gastrulation movements can range from the involution of an epithelial sheet of cells to the ingression of individual mesenchymal cells and can occur in flat or spherical tissues. Strikingly, manipulations of cell-cell interactions or of embryo geometry can change one gastrulation mode into another. These experiments suggest that tissues can easily access different morphological trajectories and that the evolution of morphogenesis might involve only minor changes in cell-cell interactions, cell shape, or initial conditions.

It will be a major challenge for the future to reconstruct the ancestral steps that have led to the combined diversification of genes, cells, and shapes that gives rise to novel phenotypes, but new technologies promise new insights. For example, comparative single-cell genomics and the systematic generation of atlases can help define cell type diversity and its underlying genetic basis.[354] For example, novel combinations of different gene expression programs (or gene modules or operons) can diversify cell types, and the divergence of paralogues after gene or genome duplication can result in cellular novelties. Spatial transcriptomics approaches will be particularly powerful because they extend comparisons from genomes and cell types to spatial patterns. For example, Tomo-seq was used to study the expression of genes that in most bilateria are expressed along the anterior-posterior axis. Strikingly, in echinoderms, which have a pentaradial body plan, these genes are expressed along the medial-lateral axis in each ray.[355]

We also foresee that the field of "developmental paleobiology" might recreate ancient cell types and organs and reconstruct the evolution of their development *in vitro*. Evo-devo combined with paleontology and genomics has produced a comprehensive understanding of the evolution of turtle shells, dinosaur skulls, and insect wings. In the future, inspiration might come from the field of biochemical paleobiology, which has resurrected ancient proteins using sequence phylogeny. Analogously, developmental paleobiology might build on the progress in tissue and organ engineering and in synthetic developmental biology.

HUMAN DEVELOPMENT

Human embryogenesis begins at fertilization and forms the major organs and tissues in the first 8 weeks of development. The fetal stage lasts from 9 weeks to birth, during which growth and maturation of body systems occurs, including the nervous system, musculoskeletal system, circulatory system, digestive system, and urogenital system. Human development differs in many respects compared to other species due to our large brains, bipedalism, and extended juvenile stages. Humans also devote a greater proportion of early development to brain growth, which constitutes most of fetal growth. Moreover, while the brains of most mammals reach adult size by the end of gestation, 80% of human brain development happens after birth over nearly two decades.

Ethical concerns prohibit the mechanistic study of these processes, but a deeper understanding of human development has been gained by the genetic analysis of birth defects and diseases. For example, mutations in Hedgehog or Nodal signaling pathways result in human midline or left-right defects, respectively, that resemble phenotypes in other vertebrate systems. In addition, mutations in developmental regulators are often associated with adult phenotypes, e.g., several malformations in the cardiovascular system are linked to mutations in genes that were identified in heart development in model organisms, and human cancers are often associated with classic developmental pathways, ranging from abnormal Wnt, Hedgehog, or

Notch signaling to cadherins, ephrins, or aberrant TF-fusions. Thus, developmental biology has been an important catalyst in biomedical research and has even laid the foundation for the development of drugs.[356]

Genomic approaches provide another way to study human development (Figure 2). For example, somatic mutations in nuclear or mitochondrial DNA have begun to generate lineage trees and clonal dynamics during human development.[357] Genome-wide association studies (GWASs) have identified polymorphisms that are associated with morphological variations. For example, craniofacial shape is influenced by hundreds of independent SNPs.[358] It has been challenging to link these SNPs to candidate genes because linkage disequilibrium hinders resolving the contribution to linked SNPs, and most GWAS loci lie in non-coding regions. However, combining GWASs with the analysis of craniofacial syndromes that are inherited at Mendelian ratios has led to the recognition that facial diversity is influenced by many TFs expressed in neural crest or mesenchymal cells during craniofacial development.[358] We expect dramatic progress in the field of human developmental variation through the advent of high-resolution multiomics atlases of human cell types, the large-scale phenotyping and genotyping of humans, and improvements in genomic and computational approaches (Box 5) that link functional variants to phenotypic changes.

Recent advances in stem cell biology and bioengineering have rapidly accelerated our understanding of human development, especially through the use of human organoids (Box 1). Organoids are 3D tissue cultures derived from stem cells (iPSCs or adult-derived SCs) that mimic developmental processes *in vitro*. For example, stem-cell-derived embryo models can partially recapitulate some of the developmental processes taking place during the first two weeks of human development. These models are incomplete, but they allow live imaging of developmental dynamics such as somitogenesis and help test the effects of teratogens or genetic perturbations. In another example, cerebral organoids contain various brain region identities and have illuminated early neural tube patterning and the consequences of perturbing cell fate. Moreover, cortical organoids can be transplanted into rat brains and develop mature cell types that integrate into sensory- and motivation-related circuits. Retinal organoids visually sensitive to light have shown how photoreceptors wire to downstream neurons. Gut organoids displaying intestinal villi and a variety of epithelial lineages have provided insight into how the human stomach and intestine emerge. Patient-derived organoids even recapitulate pathological defects[359] seen in developmental disorders, such as how autism genes converge on asynchronous development of similar classes of neurons, or in pediatric cancers such as Wilms tumors. Notably, patient-derived organoids are currently used in clinics to decide on the best treatment for cystic fibrosis patients. Tissue-derived organoids also allow us to reach frontiers in human health that were previously inaccessible. For example, endometrial organoids allow the exploration of physiological processes including the menstrual cycle and maternal-fetal interactions as well as pathological states such as endometriosis and endometrial cancer. Despite the still very rudimentary nature of currently available human organoids, these examples illustrate their power in studying human development and disease. While studies of morphogenesis are challenging due to the variability in organoid models,[360] cell fate specification and differentiation can be studied in remarkable detail. Indeed, stem cell models and organoids will often be the only way to study human-specific aspects of developmental regulation and phenotypic variation. In particular, *cis*-regulatory elements are often poorly conserved, extensively reshuffled by structural variation, and reshaped by transposable elements, limiting the use of more distantly related species such as mice.

Beyond modeling organogenesis and diseases, organoids and stem cell embryo models have raised important ethical questions. What is the limit in advancing human development *in vitro*?[361] What are the consequences of generating eggs, sperm, and zygotes from iPSCs?[362] What are the ethical and moral concerns in generating human-animal chimeras or raising animals as sources for humanized organs? Some of these questions are met with horror, and there might be insurmountable technical hurdles in the future, but it is clear that organoids present a new frontier in studying human development, reproduction, evolution, disease, and maybe even cognition while transforming therapeutic approaches. Balancing biomedical promise and ethical concerns will remain a long-term challenge for societies, including biologists and ethicists.[363]

The ethical conundrums described above also raise fundamental questions. What makes us human? What distinguishes us from other animals? What are the evolutionary underpinnings of these differences? Humans have evolved remarkable cognitive abilities that are in part associated with increased brain size as compared to non-human primates. Genomics approaches have identified gene families that were formed through gene duplications and might underlie the emergence of larger brains in humans.[364] For example, ARHGAP11B emerged from its parent gene ARHGAP11A but has acquired a new function. While ARHGAP11A is involved in cytokinesis, ARHGAP11B seems to be involved in metabolism and increases progenitor proliferation during cortical development. In another example, the human-specific NOTCH2 isoform NOTCH2NLB promotes proliferation of cortical progenitors. These examples demonstrate that genetic differences can underlie differences between humans and other primates.

Finally, what about the differences between modern-day humans and archaic humans such as Neanderthals?[365] For example, amino acid substitutions in the enzyme transketolase-like 1 have been implicated in different rates of neurogenesis between humans and Neanderthals.[366,367] However, it remains generally unclear as to which changes contributed to differences in the evolution and development between humans and Neanderthals or how environmental influences shaped their phenotypes. We foresee much progress in this field through the study of human, Neanderthal, and human-Neanderthal hybrid organoids.[365]

PROSPECTS

Our Review illustrates how the application of revolutionary technologies to developmental biology has led to fundamental insights and conceptual advances and how developmental biologists have invented powerful technologies that impact all of biomedical research. The interplay between conceptual and technological advances is complex. There are clear examples where technological progress has laid the foundation for conceptual advances (e.g., molecular cloning allowed the isolation of developmental genes discovered through genetics). Conversely, conceptual advances have led to the need for and development of new technologies (e.g., RNA *in situ* hybridization was needed to visualize the expression of cloned genes). In some cases, conceptual advances have not relied on technological advances, particularly in the field of theoretical developmental biology (e.g., somite clock and Turing patterns), but they needed to be confirmed and refined through experimental advances. The interplay between hypothesis-driven and more descriptive approaches is also complex. Several of the technologies that we introduced here—single-cell genomics, spatial transcriptomics, genetic screens, GWASs, and imaging—result in large descriptive datasets such as atlases or mutant collections that by themselves might not provide novel mechanistic insights. Hence, there has been a concern that large-scale data collection overshadows hypothesis-based curiosity-driven approaches. We would argue, however, that this perceived tension can be powerful and result in positive feedback, where new datasets inspire new questions and old questions inspire the collection of new data. Global approaches can also help to generalize conclusions that are based on anecdotal (or paradigmatic) findings. For example, early work on a handful of enhancers provided fundamental insights into the regulatory logic of patterned gene expression, but large-scale approaches generalized these findings and revealed an astounding number and diversity of enhancer elements.[182] Global approaches can also help discover patterns that were not apparent from smaller-scale approaches. For example, systematic mutagenesis studies revealed the logic of developmental pattern formation, and large-scale single-cell genomics revealed the diversity of cell types involved in processes such as lung development or immune system differentiation. We are optimistic that diverse approaches and close collaborations between groups centered in fields ranging from engineering and physics to cell and developmental biology will continue to catalyze progress in the field.

The dramatic advances in the field of AI will undoubtedly change the study of developmental biology. These tools might help address the challenge of the seemingly overwhelming complexity in developmental systems and support the integration of diverse multi-scale datasets to generate virtual representations of development. There are already numerous practical applications for AI in image analysis (Box 5), genomics,[186,190,191] and engineering.[368] AI as a support function or co-pilot will speed up tedious analyses, but it might also serve as an oracle and give us answers that we need to interpret, much like a shaman. In this incarnation, we need to be worried that AI provides biased, unreliable, false, obvious, or lowest-common-denominator answers, when in fact many of the major discoveries in developmental biology were made by noticing unusual phenomena—e.g., would AI have discovered X inactivation, microRNAs, or pair-rule genes? Should we expect an AlphaFold moment in developmental biology—a transformative advance that revolutionizes the field overnight, as happened in structural biology and might happen in cell biology?[175] For example, might it be possible to deduce the shape and composition of an organism based solely on genome sequence and initial conditions such as the shape and composition of the egg? Or do such efforts require additional information, such as how tissue-scale mechanics emerge from individual cells? Will there be a time when we can perform *in silico* mutagenesis and predict the mutant phenotypes of any organism through "virtual development?" Can the combination of AI with synthetic developmental biology create tissues or organisms with novel and compatible features? Whatever the eventual answers, developmental biology is entering a challenging and transformative phase.

While we believe that many of the concepts and ideas summarized here will stand the test of time, the continued revolution in technological approaches and their application to developmental biology will make many of our current views seem naive or obsolete. Moreover, it is very likely that there are current discoveries that will only be recognized as fundamental concepts in the future. Conversely, the future scenarios of rejuvenated humans or designer organisms might appear unethical or impossible, and the speculations that developmental biology will make key contributions to the fields of climate change and human biomedicine might be overly optimistic. However, a century ago, at the time of the Spemann-Mangold organizer experiment,[212] postulating the treatment of disease by genome-modified stem cells or predicting the generation of humans from *in vitro* fertilization[49] would have been met with disbelief and outrage. It will be thrilling to witness the next great surprises and breakthroughs in developmental biology.

SUPPLEMENTAL INFORMATION

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DECLARATION OF INTERESTS

The authors declare no competing interests.

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